Learning-Augmented Online Algorithms

Final Project for CS395T – Approximation Algorithms and Complexity

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1 Introduction

In the typical setting of online algorithms, we are tasked with tackling a problem where the input is revealed incrementally over a sequence of rounds, and we must make irrevocable decisions on each round in order to satisfy the constraints of the problem. The key challenge to overcome is that we have no knowledge about the future, yet we wish to remain competitive with respect to an optimal offline adversary that can see the entire input in advance, and in fact we wish to be competitive in the worst case over all possible inputs. However, worst-case theoretical instances of a problem might be very far from those typically encountered in practice, and so algorithms with strong guarantees in the worst-case setting might significantly underperform in the real world compared to heuristics or machine learning models.

Towards bridging this gap, one approach that has attracted significant interest over the past few years is equipping the algorithm with *predictions* regarding the specific problem instance at hand, which are assumed to have been generated by a black-box statistical model. More specifically, in the *learning-augmented* framework – introduced by [LV18] who examined the Caching problem –, the algorithm is given an input $\mathcal{I}$ revealed over time, and a problem-specific prediction $\mathcal{P}$ that could either be given in advance or also revealed over time. The prediction $\mathcal{P}$ could, for example, be a solution to the entire instance $\mathcal{I}$, or a hint as

to the action that the algorithm should take at each timestep. Crucially, no assumptions are made regarding the accuracy of the prediction, and indeed it could be entirely misleading. Our objective in this setting is to design algorithms which, when the prediction is sufficiently accurate, achieve considerably improved guarantees compared to the worst-case setting, but simultaneously, when the prediction is inaccurate, do not perform significantly worse that the best online algorithm for the problem.

To make matters more concrete, consider a cost-minimization online problem, let $S(\mathcal{P}, \mathcal{I})$ be the cost that an online algorithm would incur by blindly following prediction $\mathcal{P}$ on instance $\mathcal{I}$, and let $\mathrm{OPT}(\mathcal{I})$ be the cost of the optimal offline solution on $\mathcal{I}$. Our objective is to design an algorithm $\mathcal{A}$ whose cost $c_{\mathcal{A}}(\mathcal{P}, \mathcal{I})$ on instance $\mathcal{I}$ given prediction $\mathcal{P}$ satisfies, for some $C, R > 0$

$$c_{\mathcal{A}}(\mathcal{P}, \mathcal{I}) \leq \min\{C \cdot S(\mathcal{P}, \mathcal{I}), R \cdot \mathrm{OPT}(\mathcal{I})\}$$

An algorithm $\mathcal{A}$ satisfying this guarantee is called *C-consistent* and *R-robust*. Ideally, we would like R to be close the competitive ratio of the best online algorithm for our problem, so that even when $\mathcal{P}$ is inaccurate our performance does not suffer significantly. On the other hand, we also want $C \ll R$, so that when $\mathcal{P}$ is accurate and $S(\mathcal{P}, \mathcal{I}) \approx \mathrm{OPT}(\mathcal{I})$, our guarantee considerably improves over the offline optimal.

In some settings we will consider, our algorithm might also be given a *robustness parameter* $\lambda \in [0, 1]$, signifying the extent to which we trust the prediction $\mathcal{P}$; lower values indicate higher trust in the prediction. In that case, an algorithm $\mathcal{A}$ is called *$C(\lambda)$-consistent* and *$R(\lambda)$-robust* for some functions $C, R \colon [0, 1] \to \mathbb{R}$ if, for any instance $\mathcal{I}$, prediction $\mathcal{P}$ and robustness parameter λ, its cost is upper bounded as

$$c_{\mathcal{A}}(\mathcal{P}, \mathcal{I}, \lambda) \leq \min\{C(\lambda) \cdot S(\mathcal{P}, \mathcal{I}), R(\lambda) \cdot \mathrm{OPT}(\mathcal{I})\}$$

Naturally, when the prediction is accurate and we have high confidence in it, intuitively we would like to just follow the prediction and pay cost close to $S(\mathcal{P}, \mathcal{I})$, thus we would like $C(\lambda)$ to approach 1 for small values of λ. Conversely, when we do not trust the prediction, intuitively we would like to just ignore it and run the best online algorithm for our problem, so we would like $R(\lambda)$ to approach the competitive ratio of that algorithm for values of λ close to 1.

In this project, we will examine recent work on various classical online problems in the learning-augmented framework. For each problem we study, we will present consistent and robust algorithms, each of which accepts as input some problem-specific prediction. We will formally analyze all algorithms we consider, and provide intuition on them. Our goal is to examine the various techniques used in the design of learning-augmented algorithms, demonstrate the wide applicability of this novel framework, and identify promising open research directions in the area.

2 Ski Rental Problem

We begin by studying a simple and well-understood online problem, the Ski Rental problem. In this setting, our instance consists of T days, and we do not know T in advance. On each day, we must decide whether to rent skis for the day and incur a cost of 1, or buy skis at a known cost $B > 1$, in which case we do not incur further cost for the rest of the instance. Crucially, we do not know which the final day is until *after* it has passed, and this lack of information about the future is the main challenge we contend with. Indeed, if T is known in advance, then trivially the optimal solution is to rent every day if $T < B$ or buy on the first day if $T \geq B$. That is, the offline optimal solution for an instance $\mathcal{I}$ of Ski Rental pays cost

$$\mathrm{OPT}(\mathcal{I}) = \min(T, B)$$

In the classical online setting, it is well-known that simple randomized algorithms can achieve a competitive ratio of $\frac{e}{e-1}$, and this is optimal. Of particular interest to us is a primal-dual algorithm due to [BJN07]. The typical LP relaxation for Ski Rental, and its corresponding dual, are given as follows

Primal			Dual		
min	$B \cdot x + \sum_{t \in [T]} r_t$		max	$\sum_{t \in [T]} y_t$	
s.t.	$x + r_t \geq 1$	$\forall t \in [T]$	s.t.	$\sum_{t \in [T]} y_t \leq B$	
	$x, r_t \geq 0$	$\forall t \in [T]$		$0 \leq y_t \leq 1$	$\forall t \in [T]$

In an integral solution to the primal LP, $x = 1$ indicates that we buy skis, $r_t = 1$ indicates that we rent on day t, and the constraint enforces that we must either buy skis or rent on each day. The primal-dual algorithm of [BJN07] for this problem maintains fractional values of the primal and dual variables, which are updated on each day as in Algorithm 2.1, where we define $e(z) \coloneqq \left(1 + \frac{1}{B}\right)^{z \cdot B}$. Note that, if $B \to \infty$, then $e(z) \to e^z$, hence the choice of notation. To obtain an integral solution, we simply pick a uniformly random $u \in (0, 1)$ and buy on the first timestep t where x exceeds u. Hence, Algorithm 2.1 essentially starts with a small probability of buying and exponentially increases it at an appropriate rate on each day. In doing so we ensure that, on expectation, when we choose to buy, the total rental cost we have incurred so far is comparable to the buying cost, thus controlling our competitive ratio.

Algorithm 2.1: Ski Rental Primal-Dual

Input: Buy cost B

Initialize $x \leftarrow 0$, $r_t \leftarrow 0$ for all $t \in [T]$
for each new day t s.t. $x + r_t < 1$ **do**
  `// Primal Update`
  $r_t \leftarrow 1 - x$
  $x \leftarrow \left(1 + \frac{1}{B}\right)x + \frac{1}{e(1) \cdot B}$
  `// Dual Update`
  $y_t \leftarrow 1$

In the learning-augmented framework, we will examine the approach of [BMS20] for Ski Rental. Here, we are additionally given as input a prediction P for the total number of days, as well as a robustness parameter $\lambda \in (0, 1)$. The algorithm of [BMS20] for maintaining a fractional solution in this setting is given in Algorithm 2.2, and it is a natural extension of Algorithm 2.1; indeed, note that in the case $\lambda = 1$ (i.e. when we do not trust the prediction at all), Algorithm 2.2 recovers Algorithm 2.1. To obtain intuition on Algorithm 2.2, observe that lower values of the parameter c lead to more aggressive increases of x, which represents our probability of buying on each day. When we trust our prediction and λ is small, we want to increase x more quickly when the prediction suggests buying (i.e. when $P \geq B$) and more slowly when the prediction suggest renting (i.e. when $P < B$). And indeed, when λ is small, Algorithm 2.2 sets c to be small when $P \geq B$ and large otherwise, leading to more and less aggressive increases of x respectively.

Algorithm 2.2: Ski Rental Learning-Augmented Primal-Dual

Input: Buy cost B, prediction P, robustness parameter λ

Initialize $x \leftarrow 0$, $r_t \leftarrow 0$ for all $t \in [T]$
if $P \geq B$ **then**
  Set $c \leftarrow e(\lambda)$, $c' \leftarrow 1$
else
  Set $c \leftarrow e(1/\lambda)$, $c' \leftarrow \lambda$
for each new day t s.t. $x + r_t < 1$ **do**
  `// Primal Update`
  $r_t \leftarrow 1 - x$
  $x \leftarrow \left(1 + \frac{1}{B}\right)x + \frac{1}{(c-1) \cdot B}$
  `// Dual Update`
  $y_t \leftarrow c'$

We begin by analyzing the cost of the fractional solution constructed by Algorithm 2.2. To do so, we will first show that the algorithm constructs feasible primal and dual solutions. In fact, primal feasibility is immediate; the primal update guarantees that $x + r_t \geq 1$ whenever a new day starts, and this remains true throughout the execution since x is monotonically increasing. Thus, we turn to proving dual feasibility. To simplify the analysis, we will assume in all that follows that if $P \geq B$ then λB is an integer, otherwise B/λ is an integer; since λ is a user-defined parameter, we can always tune it to round to the nearest integer, and the analysis does not change qualitatively if we assume that we have done so.

Lemma 2.1. *Algorithm 2.2 maintains a feasible dual solution y throughout its execution.*

Proof. Observe that the only dual constraint is $\sum_{t \in [T]} y_t \leq B$, which initially holds trivially. Also, once $x \geq 1$ in the execution of the algorithm, then no more dual variables are increased, so in order to bound the total increase in $\sum y_t$ it suffices to bound the number of primal updates until $x \geq 1$. First, consider the case where $P \geq B$, so $\sum y_t$ increases by 1 on each update. Let x_t be the value of x after t primal updates, so that $x_0 = 0$. We claim that $x_t \geq \frac{e(t/B)-1}{e(\lambda)-1}$, and we show this by induction. This holds trivially for $t = 0$, so assume it holds for t, and we have by the description of the primal update

$$\begin{aligned} x_{t+1} &= \left(1 + \frac{1}{B}\right) \cdot x_t + \frac{1}{(e(\lambda) - 1) \cdot B} \\ &\geq \left(1 + \frac{1}{B}\right) \cdot \frac{e(t/B) - 1}{e(\lambda) - 1} + \frac{1/B}{e(\lambda) - 1} \\ &= \frac{(1 + 1/B) \cdot e(t/B) - 1}{e(\lambda) - 1} \\ &= \frac{e\left(\frac{t+1}{B}\right) - 1}{e(\lambda) - 1} \end{aligned}$$

where the second step is by the inductive hypothesis and the last is by definition of $e(z)$. This concludes the induction, so we indeed have $x_t \geq \frac{e(t/B)-1}{e(\lambda)-1}$ after t primal updates. Therefore, after $\lceil \lambda B \rceil \leq B$ updates, we will have $x_t \geq 1$ and we will perform no more updates. Since $\sum y_t$ increases by 1 on each update, it follows that $\sum y_t \leq B$, as required.

For the case where $P < B$, then we can simply substitute $\lambda \leftarrow 1/\lambda$ in the above inductive argument, and use the description of the primal update when $P < B$, to obtain that $x_t \geq \frac{e(t/B)-1}{e(1/\lambda)-1}$. Similarly to before, this implies that we will perform at most $\lceil B/\lambda \rceil$ updates, and we have assumed that B/λ is an integer, so we will perform at most B/λ updates. In this case, each update increases $\sum y_t$ by λ, so again we have $\sum y_t \leq B$, and the lemma follows. □

Having established primal and dual feasibility, we now turn to analyzing the consistency and robustness of our algorithm. As we argued before, the optimal offline solution on instance $\mathcal{I}$ incurs cost $\mathrm{OPT}(\mathcal{I}) = \min(T, B)$. On the other hand, given prediction P, if we were to blindly follow the prediction, we would buy on the first day if $P \geq B$ and rent every day otherwise, thus our cost would be

$$S(P, \mathcal{I}) = B \cdot \mathbb{1}\{P \geq B\} + T \cdot \mathbb{1}\{P < B\}$$

Given our above definitions for $\mathrm{OPT}(\mathcal{I})$ and $S(P, \mathcal{I})$, we have the following guarantee for the performance of the algorithm.

Lemma 2.2. *For any $\lambda \in (0, 1]$ and prediction P, the cost of the fractional solution of Algorithm 2.2 on instance $\mathcal{I}$ is at most*

$$\min\left\{\frac{\lambda}{1 - e(-\lambda)} \cdot S(P, \mathcal{I}), \frac{1}{1 - e(-\lambda)} \cdot \mathrm{OPT}(\mathcal{I})\right\}$$

Proof. We first prove the robustness bound, for which we will use an argument analogous to that of [BJN07]. By Lemma 2.1, we have established that the fractional solution satisfies primal and dual feasibility, so it

suffices to bound the per iteration increase in primal and dual objective value. Let x_t be the value of x after t primal updates, so that $x_0 = 0$. The increase in primal value during the t-th update is

$$B \cdot (x_t - x_{t-1}) + r_t = B \cdot \left(\frac{x_{t-1}}{B} + \frac{1}{(c-1)B} \right) + 1 - x_{t-1} = \frac{1}{1 - 1/c}$$

while the increase in dual value is c'. Therefore, in the case $P \geq B$ where $c = e(\lambda)$ and $c' = 1$, the ratio of primal over dual value increase is

$$\frac{1}{1 - e(-\lambda)}$$

while in the case $P < B$ where $c = e(1/\lambda)$ and $c' = \lambda$ the ratio is

$$\frac{1/\lambda}{1 - e(-1/\lambda)} \leq \frac{1}{1 - e(-\lambda)}$$

where this last inequality can be shown graphically or analytically. Therefore, in both cases the primal value of the final fractional solution is at most $\frac{1}{1-e(-\lambda)}$ times the dual value, and the robustness bound follows by weak duality (since the value of a feasible dual solution lower-bounds the optimal primal cost).

Moving on to the consistency bound, in the case $P < B$, then the prediction will pay $S(P, \mathcal{I}) = T$, while the algorithm will clearly perform at most T updates, each of which increases the primal value by $\frac{1}{1-e(-1/\lambda)}$ as we argued above. Hence, the final primal value is at most $\frac{1}{1-e(-1/\lambda)} \cdot S(P, \mathcal{I}) \leq \frac{\lambda}{1-e(-\lambda)} \cdot S(P, \mathcal{I})$, as desired. For the case $P \geq B$, recall that in the proof of Lemma 2.1 we showed that we will perform at most $\lceil \lambda B \rceil$ updates, each at a cost of $\frac{1}{1-e(-\lambda)}$ as we argued above, so the final primal value will be at most $\frac{\lceil \lambda B \rceil}{1-e(-\lambda)}$. By our assumption that λB is integral, we get that the final primal value will be at most $\frac{\lambda B}{1-e(-\lambda)}$, and since the prediction will pay $S(P, \mathcal{I}) = B$ in this case, our consistency bound again follows. □

Lemma 2.2 shows that Algorithm 2.2 maintains a feasible primal fractional solution which satisfies consistency and robustness. In order to obtain an end-to-end algorithm for Ski Rental, we need to additionally round this solution in an online fashion. This is done precisely as in [BJN07]; namely, at the beginning of the execution, we draw $u \in (0, 1)$ uniformly at random, and then choose to buy on the first day when the value of x exceeds u. If x_t is the value of x at the beginning of day t, then clearly the probability of renting on this day is $1 - x_t$, which is precisely the value of z_t in the fractional solution returned by Algorithm 2.2. Furthermore, if x_T is the value of x at the beginning of the last day, then the probability of buying at any point during the execution is exactly x_T. Since the cost of buying is R and the cost of renting is 1, it follows that the expected cost of our rounding is $B \cdot x_T + \sum_t r_t$, which is precisely the cost of the fractional solution returned by Algorithm 2.2. Thus, we can losslessly round online any solution returned by Algorithm 2.2, and this fact combined with Lemma 2.2 yields our final result for Ski Rental.

Theorem 2.3. *There exists a randomized algorithm for the Ski Rental problem that takes as input a prediction $P \in \mathbb{N}$ and a robustness parameter $\lambda \in [0, 1]$ and, on any instance $\mathcal{I}$ of ski rental, incurs expected cost at most*

$$\mathbb{E}[c(P, \mathcal{I}, \lambda)] \leq \min\left\{ \frac{\lambda}{1 - e(-\lambda)} \cdot S(P, \mathcal{I}), \frac{1}{1 - e(-\lambda)} \cdot \mathrm{OPT}(\mathcal{I}) \right\}$$

Ski Rental is a relatively straightforward and well-understood problem, so the results presented in this section are not particularly deep. However, the simplicity of this setting allows us to highlight a few points that will be recurring themes when we examine more complex problems later on. For one, observe that our approach was to begin with an optimal online algorithm for our problem and modify it to incorporate the given prediction, in a way that interpolates between fully trusting and completely ignoring the prediction. This is a widely applicable and natural methodology in the design of learning-augmented algorithms, given that we wish to achieve performance comparable to that of the best online algorithm. Secondly, observe that for small values of λ (i.e. when we trust the prediction) our consistency bound approaches 1, and so if the prediction is accurate (i.e. if the total number of days is close to P) we obtain a competitive ratio close to 1, which significantly improves over the $\frac{e}{e-1}$ optimal achievable without the use of predictions. At the

same time, for large values of λ (i.e. when we do not trust the prediction), our robustness bound approaches precisely $\frac{e}{e-1}$, and we recover the optimal online performance *regardless* of how poor our predictions were. This best-of-both-worlds tradeoff is fundamental to the usefulness and appeal of the learning-augmented framework, and it is accentuated in more challenging settings, such as those we will examine later. Indeed, we will study problems where the optimal online algorithm is unable to achieve any constant competitive ratio, yet we will be able to design learning-augmented algorithms which, when the prediction accurately captures the optimal offline solution, leverage that prediction to achieve constant-competitiveness.

3 Online Set Cover

We now turn our attention to a substantially richer classical online problem, namely Online (Weighted) Set Cover. In an instance of this problem, we are presented upfront with a universe $\mathcal{U}$ of n elements, a family $\mathcal{F} \subseteq 2^{\mathcal{U}}$ of m sets over this universe, and a weight function $w\colon \mathcal{F} \to \mathbb{R}_{\geq 0}$. We initially start with an empty solution $\mathcal{C} \subseteq \mathcal{F}$, and the instance unfolds over $T \leq n$ rounds. On each round, a new element of $\mathcal{U}$ arrives, and if our current solution $\mathcal{C}$ does not contain a set covering that element, we must immediately include such a set in $\mathcal{C}$. We are not allowed to remove sets from our solution at any point, and our objective is to minimize the total weight of the chosen sets. Throughout, we will assume that the minimum set weight satisfies $\min_{S \in \mathcal{F}} w_S \geq 1$. We can easily ensure this by rescaling the weights, which would multiply the cost of any solution by the same factor, so it does not affect our competitive analysis and is thus without loss of generality.

A first observation is that Online Set Cover generalizes the Ski Rental problem we presented above; indeed, an instance of Ski Rental is equivalent to an instance of Online Set Cover with $m = n + 1$ sets, where one set has weight B and covers all elements, while each other set has weight 1 and covers only one element, such that all single-element sets are disjoint. Again, a main challenge to overcome is uncertainty about the future, and specifically the fact that we do not know in advance which elements of $\mathcal{U}$ will arrive. However, Online Set Cover is more interesting because, unlike Ski Rental, it remains a hard problem even if we *did* know which elements would arrive, i.e. in the offline setting – specifically (offline) Set Cover is known to be NP-hard to approximate within any factor better than $\log n$.

To approach the problem, we begin by stating the typical LP relaxation for Online Set Cover, as well as the corresponding dual program. In what follows, we denote by $\mathcal{F}(e)$ the collection of sets that cover element e, i.e. $\mathcal{F}(e) \coloneqq \{S \in \mathcal{F} \mid e \in S\}$. Also, we let $e_t \in \mathcal{U}$ be the element that appears on timestep t. Naturally, each variable x_S in the primal LP represents the extent to which we include set S to our solution.

Primal			Dual		
min	$\sum_{S \in \mathcal{F}} w_S x_S$		max	$\sum_{t \in [T]} y_{e_t}$	
s.t.	$\sum_{S \in \mathcal{F}(e_t)} x_S \geq 1$	$\forall t \in [T]$	s.t.	$\sum_{t\colon e_t \in S} y_{e_t} \leq w_S$	$\forall S \in \mathcal{F}$
	$x_S \geq 0$	$\forall S \in \mathcal{F}$		$y_{e_t} \geq 0$	$\forall t \in [T]$

Online Set Cover was introduced by [AAABN09], who gave a primal-dual algorithm achieving a provably optimal competitive ratio of $O(\log n \log d)$, where d is the maximum *degree* (or *frequency*) among elements of $\mathcal{U}$, i.e. $d \coloneqq \max_{e \in \mathcal{U}} |\mathcal{F}(e)|$. That algorithm maintains, throughout its execution, fractional values for the primal and dual variables, which are updated whenever an element arrives as in Algorithm 3.1. These values are rounded in an online fashion in order to obtain an integral solution, as we will examine later. Note that the $O(\log d)$ factor in the competitive ratio is precisely due to this online rounding, while the $O(\log n)$ factor corresponds to the integrality gap of the LP, and in fact captures the hardness of approximating Set Cover even in the offline setting.

Algorithm 3.1: Online Set Cover Primal-Dual

Input: Universe $\mathcal{U}$, family of sets $\mathcal{F}$, weight function $w\colon \mathcal{F} \to \mathbb{R}_{\geq 0}$

Initialize $x_S \leftarrow 0$, $\forall S \in \mathcal{F}$ and $y_e \leftarrow 0$, $\forall e \in \mathcal{U}$
for each element e_t that arrives **do**
  while $\sum_{S \in \mathcal{F}(e_t)} x_S < 1$ **do**
    forall $S \in \mathcal{F}(e_t)$ **do**
      `// Primal Update`
      $x_S \leftarrow x_S\left(1 + \frac{1}{w_S}\right) + \frac{1}{w_S |\mathcal{F}(e_t)|}$
    `// Dual Update`
    $y_{e_t} \leftarrow y_{e_t} + 1$

Intuitively, in the unweighted case where $w_S = 1$ for all S, we can think of the primal values x_S maintained by the algorithm as an estimate for the "likelihood" that set S belongs in the optimal solution, given the elements that have appeared so far. For instance, the first time an element e arrives, Algorithm 3.1 will set the variables of all sets covering that element to $\frac{1}{|\mathcal{F}(e)|}$. Hence, if the element is covered by fewer sets, each set is individually more likely to be included in the optimal solution, and also, each of these sets is equally likely since we have not seen any other elements (and thus have no information to distinguish between the sets). Furthermore, observe that the primal updates are more aggressive when x_S is large, indicating that Algorithm 3.1 is more conservative when we have only seen a few elements belonging to a set S, however as we see more elements the rate of increase of x_S grows exponentially, as it becomes more apparent that we should include S in our solution.

In order to approach the problem in the learning-augmented setting, we will build upon Algorithm 3.1, following [BMS20]. The prediction $\mathcal{P} \subseteq \mathcal{F}$ we are given in this case is a collection of sets, intended to represent a guess for the offline optimal solution (although we do not require that $\mathcal{P}$ covers all elements that arrive). We also consider that we are given a robustness parameter $\lambda \in (0, 1)$ that quantifies our confidence in the prediction. In order to update the fractional LP variables, [BMS20] give Algorithm 3.2, which is a natural modification of Algorithm 3.1. Whenever an element e_t arrives, if it is not covered by the prediction $\mathcal{P}$, then the primal variables are simply updated as if we did not have the prediction, exactly as in Algorithm 3.1. If $\mathcal{P}$ does cover e_t, then the primal variables x_S are updated more aggressively for sets S that are part of our prediction, and this aggressiveness is controlled by how much we trust the predictions, i.e. by the robustness parameter λ. Indeed, when $\lambda = 1$ we recover Algorithm 3.1, while if $\lambda = 0$ we fully trust the prediction and only increase the dual variables for sets in it.

Algorithm 3.2: Online Set Cover Learning-Augmented Primal-Dual

Input: Universe $\mathcal{U}$, family of sets $\mathcal{F}$, weights $w\colon \mathcal{F} \to \mathbb{R}_{\geq 0}$, prediction $\mathcal{P} \subseteq \mathcal{F}$, robustness $\lambda \in (0, 1)$

Initialize $x_S \leftarrow 0$, $\forall S \in \mathcal{F}$ and $y_e \leftarrow 0$, $\forall e \in \mathcal{U}$
for each element e_t that arrives **do**
  while $\sum_{S \in \mathcal{F}(e_t)} x_S < 1$ **do**
    forall $S \in \mathcal{F}(e_t)$ **do**
      `// Primal Update`
      if $|\mathcal{F}(e_t) \cap \mathcal{P}| \geq 1$ **then**
        $x_S \leftarrow x_S\left(1 + \frac{1}{w_S}\right) + \frac{\lambda}{w_S |\mathcal{F}(e_t)|} + \frac{(1-\lambda) \cdot \mathbb{1}\{S \in \mathcal{P}\}}{w_S |\mathcal{F}(e_t) \cap \mathcal{P}|}$
      else
        `//` e_t `not covered by` $\mathcal{P}$
        $x_S \leftarrow x_S\left(1 + \frac{1}{w_S}\right) + \frac{1}{w_S |\mathcal{F}(e_t)|}$
    `// Dual Update`
    $y_{e_t} \leftarrow y_{e_t} + 1$

To analyze Algorithm 3.2, we first argue about the feasibility of the final primal-dual solution that it constructs. Primal feasibility is immediate, since, whenever an element e_t arrives, primal variables are repeatedly

increased until the corresponding primal constraint $\sum_{S \in \mathcal{F}(e_t)} x_S \geq 1$ becomes satisfied. Unlike in the case of Ski Rental, the dual solution will not necessarily be feasible, however we will now show that it becomes feasible after rescaling.

Lemma 3.1. *Let $y \in \mathbb{R}^T$ be the final dual solution constructed by Algorithm 3.2. Then for some $C = \Theta(\log d/\lambda)$, the solution $\frac{1}{C} \cdot y$ is feasible for the dual problem.*

Proof. The only dual constraints that can be violated are of the form $\sum_{e_t \in S} y_{e_t} \leq w_S$ for some $S \in \mathcal{F}$, so we fix some $S \in \mathcal{F}$ and it suffices to upper bound $\sum_{e_t \in S} y_{e_t}$. Notice that, whenever x_S is increased in the primal update, this is due to the arrival of some $e_\tau \in S$, thus y_{e_τ} is increased by 1 during the dual update and so the sum $\sum_{e_t \in S} y_{e_t}$ is also increased by 1. We will use this observation to argue that

$$x_S \geq \frac{\lambda}{d}\left(\left(1 + \frac{1}{w_S}\right)^{\sum_{e_t \in S} y_{e_t}} - 1\right)$$

which will yield an upper bound to the desired sum in terms of the primal variable x_S. We will show the above by induction on the number of primal updates on x_S. Initially, when no updates have been performed, the above holds trivially since $x_S = 0$ and $y_{e_t} = 0$ for all t, so inductively assume that it holds after k updates of x_S. By our previous observation, it follows that $\sum_{e_t \in S} y_{e_t} = k$ at this point. Consider, now, the $(k+1)$-th primal update on x_S, which is caused by the arrival of some element $e_\tau \in S$. Let x_S^k be the value of x_S before the update and x_S^{k+1} the value after the update, and similarly define y_e^k and y_e^{k+1} for any $e \in S$. Again by our previous observation, it holds that $y_{e_\tau}^{k+1} = y_{e_\tau}^k + 1$ and $y_{e_t}^{k+1} = y_{e_t}^k$ for any $t \neq \tau$. Then, regardless of which of the two cases our primal update falls under, it holds that

$$\begin{aligned}
x_S^{k+1} &\geq x_S^k\left(1 + \frac{1}{w_S}\right) + \min\left\{\frac{\lambda}{w_S|\mathcal{F}(e_\tau)|} + \frac{(1-\lambda)\cdot \mathbb{1}\{S \in \mathcal{P}\}}{w_S|\mathcal{F}(e_\tau) \cap \mathcal{P}|},\ \frac{1}{w_S|\mathcal{F}(e_\tau)|}\right\} \\
&\geq x_S^k\left(1 + \frac{1}{w_S}\right) + \frac{\lambda}{d \cdot w_S} \\
&\geq \frac{\lambda}{d}\left(\left(1 + \frac{1}{w_S}\right)^{y_{e_\tau}^k + \sum_{e_t \in S \setminus \{e_\tau\}} y_{e_t}^k} - 1\right) \cdot \left(1 + \frac{1}{w_S}\right) + \frac{\lambda}{d \cdot w_S} \\
&= \frac{\lambda}{d}\left(\left(1 + \frac{1}{w_S}\right)^{y_{e_\tau}^{k+1} + \sum_{e_t \in S \setminus \{e_\tau\}} y_{e_t}^k} - 1\right) \\
&= \frac{\lambda}{d}\left(\left(1 + \frac{1}{w_S}\right)^{\sum_{e_t \in S} y_{e_t}^{k+1}} - 1\right)
\end{aligned}$$

The second line is because $\lambda \leq 1$ and $|\mathcal{F}(e_\tau)| \leq d$ by definition of the maximum degree d, the third is by the inductive hypothesis, while the last two are by our previous relations between $y_{e_t}^k$ and $y_{e_t}^{k+1}$. This concludes our inductive argument, hence we have shown that the final dual solution constructed by the algorithm satisfies

$$\begin{aligned}
x_S &\geq \frac{\lambda}{d}\left(\left(1 + \frac{1}{w_S}\right)^{\sum_{e_t \in S} y_{e_t}} - 1\right) \\
&= \frac{\lambda}{d}\left(\left(1 + \frac{1}{w_S}\right)^{w_S \cdot \frac{\sum_{e_t \in S} y_{e_t}}{w_S}} - 1\right) \\
&\geq \frac{\lambda}{d} \cdot \left(2^{\frac{\sum_{e_t \in S} y_{e_t}}{w_S}} - 1\right)
\end{aligned}$$

where the last line is because we have assumed $w_S \geq 1$, which implies $\left(1 + \frac{1}{w_S}\right)^{w_S} \geq 2$. Rearranging, we obtain an upper bound on the desired sum

$$\sum_{e_t \in S} y_{e_t} \leq w_S \cdot \log\left(1 + \frac{x_S \cdot d}{\lambda}\right)$$

To conclude the proof, it suffices to obtain an upper bound on the final value of the variable x_S. To do so, note that once x_S becomes at least 1, no more primal updates will be performed on it. Therefore, if x'_S is the value of the primal variable right before its final update, we have $x'_S < 1$. Hence, regardless regardless of which of the two cases our primal update falls under, if e_τ is the element whose arrival caused this last update, it holds that

$$x_S \leq x'_S\left(1 + \frac{1}{w_S}\right) + \max\left\{\frac{\lambda}{w_S|\mathcal{F}(e_\tau)|} + \frac{(1-\lambda)\cdot \mathbb{1}\{S \in \mathcal{P}\}}{w_S|\mathcal{F}(e_\tau)\cap\mathcal{P}|},\ \frac{1}{w_S|\mathcal{F}(e_\tau)|}\right\} \leq 2x'_S + 1 \leq 3$$

since $w_S \geq 1$, $|\mathcal{F}(e_\tau)| \geq 1$ and $\lambda \leq 1$. Combining our last two displays yields the lemma. □

Lemma 3.1 shows that the dual solution constructed by Algorithm 3.2 is approximately feasible. Hence, in order to argue about the value of the primal solution, it essentially suffices to bound the ratio of the final cost of the primal solution to the final value of the dual solution. Concretely, let $\mathrm{OPT}(\mathcal{I})$ denote the cost of an offline optimal solution on instance $\mathcal{I}$, and we have the following lemma.

Lemma 3.2. *On any instance $\mathcal{I}$ of Online Set Cover, the cost of the (fractional) primal solution constructed by Algorithm 3.2 satisfies*

$$\sum_{S\in\mathcal{F}} w_S x_S \leq O\left(\log\frac{d}{\lambda}\right)\cdot \mathrm{OPT}(\mathcal{I})$$

Proof. Consider any iteration of the while-loop of the algorithm, i.e. any pair of primal-dual updates. We will bound the ratio of increase in primal cost over increase in dual value during this update. Let x_S, y_{e_t} be the values of the dual variables before this update, and x'_S, y'_{e_t} the corresponding values after the update. Suppose that e_τ is the element whose arrival caused this update. Then, as we observed previously, $y'_{e_\tau} = y_{e_\tau} + 1$ and $y'_{e_t} = y_{e_t}$ for any $t \neq \tau$, so the change in dual value during the update is

$$\Delta D \coloneqq \sum_t y'_{e_t} - \sum_t y_{e_t} = 1$$

For the primal update, we discern two cases, depending if the prediction $\mathcal{P}$ covers the arrived element e_τ. If it does, then the primal update gives the following change in primal cost

$$\begin{aligned}
\Delta P &\coloneqq \sum_{S\in\mathcal{F}} w_S\cdot(x'_S - x_S) = \sum_{S\in\mathcal{F}(e_\tau)} w_S\cdot(x'_S - x_S)\\
&= \sum_{S\in\mathcal{F}(e_\tau)} w_S\cdot\left(\frac{x_S}{w_S} + \frac{\lambda}{w_S|\mathcal{F}(e_t)|} + \frac{(1-\lambda)\cdot\mathbb{1}\{S\in\mathcal{P}\}}{w_S|\mathcal{F}(e_t)\cap\mathcal{P}|}\right)\\
&= \sum_{S\in\mathcal{F}(e_\tau)} x_S + \sum_{S\in\mathcal{F}(e_\tau)}\frac{\lambda}{|\mathcal{F}(e_t)|} + \sum_{S\in\mathcal{F}(e_\tau)\cap\mathcal{P}}\frac{1-\lambda}{|\mathcal{F}(e_t)\cap\mathcal{P}|}\\
&= \sum_{S\in\mathcal{F}(e_\tau)} x_S + \lambda + (1-\lambda)\\
&\leq 2
\end{aligned}$$

where the last line is because $\sum_{S\in\mathcal{F}(e_\tau)} x_S < 1$, otherwise we would not perform a primal update. Similarly, if $\mathcal{P}$ does not cover e_τ, then the change in primal cost is

$$\begin{aligned}
\Delta P &= \sum_{S\in\mathcal{F}(e_\tau)} w_S\cdot(x'_S - x_S)\\
&= \sum_{S\in\mathcal{F}(e_\tau)} x_S + \sum_{S\in\mathcal{F}(e_\tau)}\frac{1}{|\mathcal{F}(e_t)|}\\
&= 1 + \sum_{S\in\mathcal{F}(e_\tau)} x_S \leq 2
\end{aligned}$$

Therefore, on any primal-dual update, the ratio of increase in primal cost over increase in dual value is bounded by $\frac{\Delta P}{\Delta D} \leq 2$, and so the final primal and dual solutions satisfy

$$\sum_{S \in \mathcal{F}} w_S x_S \leq 2 \sum_{t \in [T]} y_{e_t}$$

Also, Lemma 3.1 and weak duality give us

$$\frac{\sum_t y_{e_t}}{O\left(\log \frac{d}{\lambda}\right)} \leq \mathrm{OPT}(\mathcal{I})$$

Combining the last two displays yields the lemma. □

The previous lemma bounds the robustness of the fractional solution returned by Algorithm 3.2. Importantly, when λ is a constant away from zero (i.e. when we do not blindly trust the prediction), the algorithm is $O(\log d)$-robust, which essentially matches the competitive ratio of Algorithm 3.1. Thus, our performance does not suffer significantly compared to the best known (fractional) online algorithm for the problem, even when the prediction is wholly inaccurate. However, when the prediction *is* accurate, we can even hope to be constant-competitive, as we will see shortly in our consistency bound.

For an instance $\mathcal{I}$ of Online Set Cover and a prediction $\mathcal{P}$, let $S(\mathcal{P}, \mathcal{I})$ be the total weight of sets in $\mathcal{P}$. If $\mathcal{P}$ is a feasible solution, i.e. if it covers all elements that arrive, then the natural consistency bound to prove is with respect to $S(\mathcal{P}, \mathcal{I})$, i.e. the cost of blindly following $\mathcal{P}$ and buying all sets in it. However, since we do not require that $\mathcal{P}$ be a feasible solution, we will prove a modified consistency bound. Let $C(\mathcal{P}, \mathcal{I})$ be the minimum cost to cover all elements that arrive in instance $\mathcal{I}$ and are not covered by $\mathcal{P}$. Then we have the following lemma.

Lemma 3.3. *On any instance $\mathcal{I}$ of Online Set Cover, for any prediction $\mathcal{P}$, the cost of the (fractional) primal solution constructed by Algorithm 3.2 satisfies*

$$\sum_{S \in \mathcal{F}} w_S x_S \leq O\left(\frac{1}{1-\lambda}\right) \cdot S(\mathcal{P}, \mathcal{I}) + O(\log d) \cdot C(\mathcal{P}, \mathcal{I})$$

Proof. Let $\mathcal{C} \subseteq \mathcal{U}$ be the set of elements that arrive in $\mathcal{I}$ and are not covered by $\mathcal{P}$, and let $y_{\mathcal{C}}$ be the vector of dual variables corresponding to those elements. Consider first all primal updates caused by the arrival of some element in $\mathcal{C}$, and note that these primal updates simply emulate the case $\lambda = 1$, where we ignore the prediction $\mathcal{P}$. By the exact same calculations as in Lemma 3.1 (changing only the way that primal variables are updated), we can show that $\frac{y_{\mathcal{C}}}{O(\log d)}$ is a feasible solution to the dual program restricted to elements in $\mathcal{C}$. Then, exactly as in Lemma 3.2, we can show that the total cost of all the primal updates caused by elements in $\mathcal{C}$ is upper bounded by $O(\log d) \cdot C(\mathcal{P}, \mathcal{I})$. Therefore, in order to show the lemma, it suffices to show that the total cost of primal updates caused by elements not in $\mathcal{C}$, i.e. elements covered by $\mathcal{P}$, is at most $O\left(\frac{1}{1-\lambda}\right) \cdot S(\mathcal{P}, \mathcal{I})$.

Consider the primal update caused by the arrival of an element e_τ covered by $\mathcal{P}$. Let $\mathcal{G} \coloneqq \mathcal{F}(e_\tau) \cap \mathcal{P}$ be the sets of $\mathcal{P}$ that cover e_τ. Then, the change in primal value during the update can be decomposed into two parts: the change due to sets in $\mathcal{G}$, denoted by $\Delta P_{\mathcal{G}}$, and the change due to sets not in $\mathcal{G}$, denoted $\Delta P_{\bar{\mathcal{G}}}$. By the description of the primal update, we have that

$$\begin{aligned}
\Delta P_{\mathcal{G}} &= \sum_{S \in \mathcal{G}} w_S \cdot (x'_S - x_S) \\
&= \sum_{S \in \mathcal{G}} w_S \cdot \left(\frac{x_S}{w_S} + \frac{\lambda}{w_S |\mathcal{F}(e_t)|} + \frac{1-\lambda}{w_S |\mathcal{F}(e_t) \cap \mathcal{P}|} \right) \\
&= \sum_{S \in \mathcal{G}} x_S + \frac{\lambda |\mathcal{G}|}{|\mathcal{F}(e_t)|} + 1 - \lambda \\
&\geq \frac{\lambda}{d} + 1 - \lambda
\end{aligned}$$

since $|\mathcal{G}| \geq 1$ (because e_τ is covered by $\mathcal{P}$) and $|\mathcal{F}(e_t)| \leq d$ by definition of d. Also, we have

$$\begin{aligned}
\Delta P_{\bar{\mathcal{G}}} &= \sum_{S \in \mathcal{F}(e_t) \setminus \mathcal{G}} w_S \cdot (x'_S - x_S) \\
&= \sum_{S \in \mathcal{F}(e_t) \setminus \mathcal{G}} w_S \cdot \left(\frac{x_S}{w_S} + \frac{\lambda}{w_S |\mathcal{F}(e_t)|} \right) \\
&= \sum_{S \in \mathcal{G}} x_S + \frac{\lambda |\mathcal{F}(e_t) \setminus \mathcal{G}|}{|\mathcal{F}(e_t)|} \\
&\leq 1 + \lambda
\end{aligned}$$

since $|\mathcal{F}(e_t) \setminus \mathcal{G}| \leq |\mathcal{F}(e_t)|$ and $\sum_{S \in \mathcal{G}} x_S \leq \sum_{S \in \mathcal{F}(e_\tau)} x_S < 1$, otherwise we would not perform a primal update. Combining the above we obtain that $\Delta P_{\bar{\mathcal{G}}} \leq \frac{1+\lambda}{\lambda/d+1-\lambda} \cdot \Delta P_{\mathcal{G}}$, and therefore the total increase in primal cost during the update is bounded by

$$\Delta P \coloneqq \Delta P_{\mathcal{G}} + \Delta P_{\bar{\mathcal{G}}} \leq \frac{2 + \lambda/d}{1 - (1 - 1/d)\lambda} \cdot \Delta P_{\mathcal{G}} \leq O\left(\frac{1}{1-\lambda} \right) \cdot \Delta P_{\mathcal{G}}$$

Therefore, the total increase in primal cost due to elements covered by $\mathcal{P}$ is within a factor of $O\left(\frac{1}{1-\lambda}\right)$ of the total increase in primal cost of sets in $\mathcal{P}$, which equals $\sum_{S \in \mathcal{P}} w_S x_S$ for the final values of the primal variables. In the proof of Lemma 3.1 we argued that those final values satisfy $x_S \leq 3$, so overall the total increase in primal cost due to elements covered by $\mathcal{P}$ is within a factor of $O\left(\frac{1}{1-\lambda}\right)$ of $\sum_{S \in \mathcal{P}} w_S$, which is precisely equal to $S(\mathcal{P}, \mathcal{I})$ by definition, so we obtain the lemma. □

Lemmas 3.2 and 3.3 establish the robustness and consistency of the fractional solution constructed by Algorithm 3.2. To obtain a complete algorithm, all that remains is to round the solution online, and we can use a simple rounding scheme as explained in [BS09]. Specifically, before elements arrive, for each set $S \in \mathcal{F}$ we draw $\Theta(\log n)$ random values uniformly in $[0, 1]$, and set a threshold τ_S as the minimum of those values. Then, during the execution of the algorithm, we include S in our solution the first time its primal variable x_S exceeds τ_S. Thus, for any set S, if the final value of its primal variable is x_S, then by a union bound the probability that we buy S (i.e. at least one of our $\Theta(\log n)$ random values are below x_S) is at most $\Theta(\log n) \cdot x_S$, so by linearity the expected cost of the sets bought is at most $\Theta(\log n) \cdot \sum_S w_S x_S$, i.e. at most $\Theta(\log n)$ times the cost of the fractional solution. Also, whenever an element e_t arrives, if x_S^t is the primal value of a set S after the primal updates, then the probability that we do not cover e_t is

$$\prod_{S \ni e_t} \left(1 - x_S^t\right)^{\Theta(\log n)} \leq \exp\left(-\Theta(\log n) \cdot \sum_{S \ni e_t} x_S^t \right) \leq n^{-\Theta(1)}$$

since, after the primal updates, it holds that $\sum_{S \ni e_t} x_S^t \geq 1$. Thus, by a union bound, all elements will be covered when they arrive with probability $1 - n^{-\Theta(1)}$. Therefore, combining this rounding scheme with the results of Lemmas 3.2 and 3.3 shows our main result for Online Set Cover in the learning augmented framework.

Theorem 3.4. *There exists a randomized algorithm for Online Set Cover which, on any instance $\mathcal{I}$ on a universe with n elements and set family $\mathcal{F}$, takes as input a prediction $\mathcal{P} \subseteq \mathcal{F}$ and a robustness parameter $\lambda \in [0, 1]$, covers all elements that arrive with probability at least $1 - n^{-\Theta(1)}$, and incurs expected cost at most*

$$\mathbb{E}[c(\mathcal{P}, \mathcal{I}, \lambda)] \leq O(\log n) \cdot \min\left\{ O\left(\frac{1}{1-\lambda} \right) \cdot S(\mathcal{P}, \mathcal{I}) + O(\log d) \cdot C(\mathcal{P}, \mathcal{I}),\ O\left(\log \frac{d}{\lambda} \right) \cdot \mathrm{OPT}(\mathcal{I}) \right\}$$

The upshot of this theorem is that, given a relatively accurate prediction $\mathcal{P}$ (i.e. a prediction with cost $S(\mathcal{P}, \mathcal{I}) \approx \mathrm{OPT}(\mathcal{I})$ that covers all but a small enough fraction of elements that arrive), we can pick some constant robustness parameter λ and obtain cost approximately $O(\log n) \cdot \mathrm{OPT}(\mathcal{I})$, which significantly improves

upon the lower bound of $\Omega(\log n \log d)$ for any online algorithm. At the same time, for any constant robustness λ, even if our prediction is completely inaccurate, we still recover the $O(\log n \log d)$-competitiveness of the algorithm of [AAABN09]. Thus, we have presented our first concrete application of the learning-augmented framework on a problem significantly richer and harder than Ski Rental, which further demonstrates the utility of the framework.

4 Online Metric Facility Location

4.1 Problem Definition and Main Result

Remaining withing the theme of online covering problems, we now examine the well-studied *Online Metric Facility Location* problem. Here, we are given a metric space $(\mathcal{X}, d)$, a ground set $\mathcal{D} \subseteq \mathcal{X}$ of demand points, a ground set $\mathcal{F} \subseteq \mathcal{X}$ of facilities and an opening cost function $w\colon \mathcal{F} \to \mathbb{R}_{>0}$. Initially, all facilities in $\mathcal{F}$ are closed, and we can open a facility $f \in \mathcal{F}$ at any time, paying its opening cost $w(f)$. The input sequence consists of n demand points $X \coloneqq (x_1, \ldots, x_n) \subseteq \mathcal{D}$ which arrive one-by-one. Whenever a demand x_i arrives, we must *irrevocably* assign it to some open facility f_i, paying connection cost equal to the distance $d(x_i, f_i)$. Once we make the assignment, we cannot change it later, and we also cannot close a facility once we open it. Our goal is to minimize the total cost incurred, i.e.

$$\sum_{f \in F} w(f) + \sum_{x_i \in X} d(x_i, f_i)$$

where F is the set of open facilities.

Online Metric Facility Location is an extensively studied problem, and it is a generalization of Online Set Cover (to see this, given an instance of Online Set Cover, we can identify sets with facilities, elements with demands, and define the distance between a demand and a facility to be 0 if the corresponding set contains the corresponding element, and ∞ otherwise). The problem was introduced by [Mey01], who proposed Algorithm 4.1 and showed that it is $O(\log n)$-competitive. In fact, [Fot08] showed that Algorithm 4.1 is in fact $O\left(\frac{\log n}{\log \log n}\right)$-competitive, and also showed that no randomized algorithm can achieve a competitive ratio better than $\Omega\left(\frac{\log n}{\log \log n}\right)$ against an oblivious adversary. Hence, Algorithm 4.1 is in fact an optimal algorithm for Online Metric Facility Location and, similarly to the previous sections, we will build upon it in the learning-augmented setting.

In the following description of Algorithm 4.1, we assume that the opening costs w are normalized, so that the minimum opening cost equals 1. We also assume that we have rounded down the opening cost of each facility to the closest power of 2. This only increases the facility cost paid by the algorithm by a factor of 2, since the actual facility costs may be up to twice the assumed costs. In other words, we assume that the domain of w is $\{2^{i-1} \mid i \in [L]\}$ for some $L \geq 1$. Finally, we denote by G_k the set of facilities whose opening cost is at most 2^{k-1}, that is, $G_k \coloneqq \{f \in \mathcal{F} \mid w(f) \leq 2^{k-1}\}$ for each $0 \leq k \leq L$, so that $G_0 = \emptyset$.

Algorithm 4.1: Meyerson's Algorithm

Input: Metric space $(\mathcal{X}, d)$, demands $\mathcal{D} \subseteq \mathcal{X}$, facilities $\mathcal{F} \subseteq \mathcal{X}$, opening costs $w\colon \mathcal{F} \to \mathbb{R}_{>0}$

Initialize the set of open facilities $F \leftarrow \emptyset$
for each demand x that arrives **do**
  for $k \in [L]$ **do**
    $f_k \leftarrow \operatorname{argmin}_{f \in F \cup G_k} d(x, f)$
    $\delta_k \leftarrow d(x, f_k)$
    $p_k \leftarrow \frac{\delta_{k-1} - \delta_k}{2^k}$, where $\delta_0 \coloneqq \min_{f \in F} d(x, f)$
  Define the partial sums $s_k \leftarrow \sum_{i=k}^{L} p_i$ for $k \in [L]$
  Draw r uniformly from $[0, 1]$
  if there exists $k \in [L]$ such that $r \in [s_{k+1}, s_k)$ **then**
    $F \leftarrow F \cup f_k$ `// Open facility` f_k
  Connect x to its nearest neighbor in F

In words, Algorithm 4.1 considers opening a new facility of each different opening cost value. For each possible opening cost 2^{k-1} for $k \in [L]$, the algorithm finds the closest facility f_k that can be opened with cost at most 2^{k-1} (where facilities that are already opened cost 0 to open), and sets δ_k to be its distance from the arrived demand. Then, each facility f_k is opened with probability $p_k = \frac{\delta_{k-1}-\delta_k}{2^k}$. The reason for choosing the opening probabilities in this manner is that it allows us to relate the cost of connecting the arrived request x (which is at most $\delta_0 \coloneqq \min_{f \in F} d(x, f)$) to the expected facility opening cost caused by the arrival of x (which is $\sum_k \frac{\delta_{k-1}-\delta_k}{2^k} \cdot 2^k \leq \delta_0$).

In the learning-augmented setting, we will assume that, whenever a demand point x arrives, we are given a predicted facility f_x^{pred} that is supposed to be the facility to which x is assigned in the optimal offline solution. If f_x^{opt} is the facility to which x is actually assigned in the optimal solution, our hope is that, if f_x^{pred} is close to f_x^{opt}, then we can leverage the predictions in order to bypass the $\Omega\left(\frac{\log n}{\log \log n}\right)$ lower bound. As we will see, we will be able to give an algorithm whose guarantee depends upon the maximum distance between $f_{x_i}^{\text{pred}}$ and $f_{x_i}^{\text{opt}}$ for all demands x_i, and so we will find the following definition useful.

Definition 4.1 (Maximum prediction error)**.** *Let $(x_1, \ldots, x_n) \subseteq \mathcal{D}$ be a sequence of demand points. For each demand x_i, let $f_{x_i}^{\text{pred}} \in \mathcal{F}$ be the predicted facility for x_i, and let $f_{x_i}^{\text{opt}} \in \mathcal{F}$ be the facility to which x_i is assigned in the offline optimal solution. The* maximum prediction error *η_∞ is defined as*

$$\eta_\infty \coloneqq \max_{1 \leq i \leq n} d(f_{x_i}^{\text{pred}}, f_{x_i}^{\text{opt}})$$

A first observation is that η_∞ can be unbounded if the predictions are bad enough. However, it is possible to "calibrate" the predictions as follows: when a demand x arrives, we compute $f'_x \coloneqq \text{argmin}_{f \in \mathcal{F}}\{d(x, f'_x) + w(f)\}$, and then, if $d(x, f_x^{\text{pred}}) \geq 2d(x, f'_x) + w(f'_x)$, we use f'_x in place of the prediction f_x^{pred}. In other words, if the predicted facility f_x^{pred} is too far away from the demand x, we simply ignore it and use a more "reasonable" facility as the prediction. In doing so, we ensure that $\eta_\infty = O(\text{OPT})$; indeed, for any demand x, we have that $d(f'_x, f_x^{\text{opt}}) \leq d(x, f'_x) + d(x, f^{\text{opt}}) \leq 2d(x, f'_x) + w(f'_x) \leq 2 \cdot \text{OPT}$, where the first step is by the triangle inequality, the second by optimality of f^{opt} (i.e. the connection cost $d(x, f_x^{\text{opt}})$ has to be smaller than the cost of first opening f'_x and then connecting x go f'_x), and the last by definition of f'_x (opening f'_x and connecting x to f'_x is the cheapest way to satisfy x). Thus, in all that follows, we will assume that we calibrate each prediction we receive, and so $\eta_\infty = O(\text{OPT})$.

We are now ready to present our algorithm for the learning-augmented case. As we did in the previous sections, in order to construct an algorithm which not only can leverage accurate predictions, but also is almost as good as the optimal online algorithm when the predictions are inaccurate, we will appropriately modify an existing optimal online algorithm, which in this case is Algorithm 4.1. In particular, we will analyze Algorithm 4.2, which was given by [JLLTZ22], and is a natural extension of Algorithm 4.1 in the following way: whenever a new demand x arrives, our first step is to run Algorithm 4.1 to potentially open a new facility, and then connect x to its closest facility. Let q is the total cost that we paid in (potentially) opening a facility and connecting x during this call. Our second step is to look at the predicted facility f_x^{pred} and open as many facilities as possible that are "close" to it, so that the total (expected) cost of facilities opened in this step is at most q. It is clear that our modified algorithm will pay at most twice the cost of simply running Algorithm 4.1 on the entire input, and thus, its performance will not be significantly worse than Algorithm 4.1, even if the predictions are entirely misleading. However, if the predictions are actually accurate, we can achieve a significantly better competitive ratio, as is captured by the following theorem.

Theorem 4.1. *There exists a randomized algorithm for Online Metric Facility Location (Algorithm 4.2) which, on any instance $\mathcal{I}$ with n demands to be satisfied, given a set of predicted facilities with maximum error η_∞, incurs expected cost at most*

$$\mathbb{E}[c(\mathcal{I})] \leq O\left(\min\left\{\log \frac{n\eta_\infty}{\text{OPT}(\mathcal{I})}, \log n\right\}\right) \cdot \text{OPT}(\mathcal{I})$$

Algorithm 4.2: Learning-Augmented Meyerson

Input: Metric space $(\mathcal{X}, d)$, demands $\mathcal{D} \subseteq \mathcal{X}$, facilities $\mathcal{F} \subseteq \mathcal{X}$, opening costs $w\colon \mathcal{F} \to \mathbb{R}_{>0}$

Procedure MAIN():
  // F is the set of all open facilities and F_P are those opened by PRED()
  Initialize $F \leftarrow \emptyset$, $F_P \leftarrow \emptyset$
  for each demand x that arrives **do**
    $q \leftarrow$ MEY(x)
    PRED(f_x^{pred}, q)

Procedure MEY(x):
  for $k \in [L]$ **do**
    $f_k \leftarrow \operatorname{argmin}_{f \in F \cup G_k} d(x, f)$
    $\delta_k \leftarrow d(x, f_k)$
    $p_k \leftarrow \frac{\delta_{k-1} - \delta_k}{2^k}$, where $\delta_0 \coloneqq d(x, F)$
  Define the partial sums $s_k \leftarrow \sum_{i=k}^{L} p_i$ for $k \in [L]$
  Draw r uniformly from $[0, 1]$
  if there exists $k \in [L]$ such that $r \in [s_{k+1}, s_k)$ **then** // Open facility f_k
    $F \leftarrow F \cup f_k$
    $q \leftarrow d(x, F) + w(f_k)$
  else
    $q \leftarrow d(x, F)$
  Connect x to its nearest neighbor in F
  return q

Procedure PRED(f^{pred}, q):
  repeat
    $r \leftarrow \frac{1}{2} d(f^{\text{pred}}, F_P)$
    $f^{\text{open}} \leftarrow \operatorname{argmin}_{f\colon d(f^{\text{pred}}, f) \le r} w(f)$ // Break ties by picking closest facility to f^{pred}
    if $q \ge w(f^{\text{open}})$ **then** // Open facility f^{open}
      $F \leftarrow F \cup f^{\text{open}}$, $F_P \leftarrow F_P \cup f^{\text{open}}$
      $q \leftarrow q - w(f^{\text{open}})$
  until $q < w(f^{\text{open}})$
  With probability $\frac{q}{w(f^{\text{open}})}$, set $F \leftarrow F \cup f^{\text{open}}$, $F_P \leftarrow F_P \cup f^{\text{open}}$

A couple of remarks are in order regarding Theorem 4.1. First, note that since the optimal solution has to serve n demands, and will typically pay at least $\Omega(1)$ for each demand, we can reasonably consider that $\text{OPT} = \Omega(n)$. Therefore, when our predictions have constant maximum error $\eta_\infty = O(1)$, Theorem 4.1 gives a *constant*-competitive algorithm for the problem, while if the predictions have arbitrary error we recover an $O(\log n)$ competitive ratio, which as we discussed is close to the online optimal. Secondly, since we assumed that the predictions are calibrated and $\eta_\infty = O(\text{OPT})$, the term $\log \frac{n\eta_\infty}{\text{OPT}(\mathcal{I})}$ is always $O(\log n)$, and so in order to prove Theorem 4.1 it suffices to show a single bound $O\left(\log \frac{n\eta_\infty}{\text{OPT}(\mathcal{I})}\right)$ for the total expected cost.

In order to analyze Algorithm 4.2 and show Theorem 4.1, we will require some notation and definitions, detailed below:

- We fix an optimal solution and denote by f_{opt} the set of open facilities in that solution. For any $f^* \in f_{\text{opt}}$, we let $X(f^*)$ be the demands assigned to f^* in this solution.
- During the execution of Algorithm 4.2, for any demand x, we let $F_b(x)$ be the set of open facilities right before the arrival of x and $F_a(x)$ be the open facilities after x has been processed. Thus, the set of facilities that are newly-opened due to the arrival of x is $F_n(x) \coloneqq F_a(x) \setminus F_b(x)$. Similarly, we let $F_b^M(x), F_a^M(x)$ be the facilities that are opened during the MEY step before and after the arrival of x, and $F_n^M(x) \coloneqq F_a^M(x) \setminus F_b^M(x)$. For the PRED step, we analogously define $F_b^P(x), F_a^P(x)$ and $F_n^P(x)$.

- For each demand x, we define the total cost incurred during the MEY step as $\text{cost}_M(x) \coloneqq w\big(F_n^M(x)\big) + d\big(x, F_a^M(x) \cup F_b^P(x)\big)$, the total cost during the PRED step as $\text{cost}_P(x) \coloneqq w\big(F_n^P(x)\big)$, and the overall total cost of the demand as $\text{cost}(x) \coloneqq \text{cost}_M(x) + \text{cost}_P(x)$.

To prove Theorem 4.1, we will rely on the following lemma, which constitutes the main technical result of this section.

Lemma 4.2. *For every facility $f^* \in f_{\text{opt}}$, it holds that*

$$\sum_{x \in X(f^*)} \mathbb{E}[\text{cost}_M(x)] \leq O\left(\log \frac{n\eta_\infty}{\text{OPT}(\mathcal{I})}\right) \cdot \left(w(f^*) + \sum_{x \in X(f^*)} d(x, f^*)\right) + O\left(\frac{|X(f^*)|}{n} \cdot \text{OPT}\right)$$

Given Lemma 4.2, it is straightforward to prove the main theorem Theorem 4.1, as follows.

Proof of Theorem 4.1. First, observe that for each demand x, the total expected cost is $\mathbb{E}[\text{cost}(x)] = \mathbb{E}[\text{cost}_M(x) + \text{cost}_P(x)] = 2\mathbb{E}[\text{cost}_M(x)]$, since clearly the expected cost of the $\textsc{Pred}(f^{\text{pred}}, q)$ step is q by definition of PRED, and also Algorithm 4.2 sets the budget q to be equal to the cost incurred during the MEY step for x, i.e. equal to $\text{cost}_M(x)$. Thus, the total cost of the algorithm is bounded as

$$\begin{aligned}
\mathbb{E}[\text{ALG}] &= \sum_{x \in X} \mathbb{E}[\text{cost}(x)] = 2 \sum_{x \in X} \mathbb{E}[\text{cost}_M(x)] = 2 \sum_{f^* \in f_{\text{opt}}} \sum_{x \in X(f^*)} \mathbb{E}[\text{cost}_M(x)] \\
&\leq \sum_{f^* \in f_{\text{opt}}} \left(O\left(\log \frac{n\eta_\infty}{\text{OPT}(\mathcal{I})}\right) \cdot \left(w(f^*) + \sum_{x \in X(f^*)} d(x, f^*)\right) + O\left(\frac{|X(f^*)|}{n} \cdot \text{OPT}\right) \right) \\
&= O\left(\log \frac{n\eta_\infty}{\text{OPT}(\mathcal{I})}\right) \cdot \text{OPT}
\end{aligned}$$

where the inequality is by Lemma 4.2. □

Therefore, we now turn our attention to proving our main lemma, and this will imply our claimed competitive ratio guarantee.

4.2 Proof of Lemma 4.2

In what follows, we fix an arbitrary facility $f^* \in f_{\text{opt}}$. We will require some further notation. Recalling that we have assumed all opening costs to be powers of 2, let ℓ^* be the integer such that $w(f^*) = 2^{\ell^* - 1}$, or equivalently $f^* \in G_{\ell^*}$ by our previous notation. Let $X(f^*) = \{x_1, \ldots, x_m\}$ be listed according to their arrival order, so that, for any $i \in [m]$, if a demand x' arrives between x_i and x_{i+1}, then it must be connected to some facility other than f^* in the optimal solution. Finally, let τ be the largest integer such that $d(f^*, F_b(x_\tau)) > \min\{7\eta_\infty, 4w(f^*)\}$. Note that τ is a random variable.

To bound the total expected cost of the MEY steps and prove Lemma 4.2, we will split the analysis into two steps: the *long-distance* stage corresponding to the demands $(x_1, \ldots, x_\tau)$ and the *short-distance* stage corresponding to $(x_{\tau+1}, \ldots, x_m)$. We begin by considering the long-distance stage.

4.2.1 Long-distance stage

In the long-distance stage, we intuitively expect that the PRED step will quickly open a facility within a distance of at most $O(\eta_\infty)$ of f^*, since, for every $x_i \in X(f^*)$, we have that $d(f_{x_i}^{\text{pred}}, f^*) \leq \eta_\infty$, and this allows us to show bounds on the cost of the long-distance stage. For the analysis, we will need to discern two cases depending on which of the two cases in the definition of τ hold, i.e. on whether $7\eta_\infty \leq 4w(f^*)$ or $7\eta_\infty > 4w(f^*)$. We begin by considering the first case.

Lemma 4.3. *Suppose that $7\eta_\infty \le 4w(f^*)$. Then it holds that*

$$\mathbb{E}\left[\sum_{i<\tau} \mathrm{cost}_M(x_i)\right] \le 2w(f^*)$$

Proof. Let $G \coloneqq \{g_1, \dots, g_t\}$ be the facilities opened during PRED steps on the arrivals of the demands $\{x_i\}_{i<\tau}$. As we observed earlier, the expected cost of each PRED step is precisely equal to the cost of its preceding MEY step, hence

$$\mathbb{E}\left[\sum_{i<\tau} \mathrm{cost}_M(x_i)\right] = \mathbb{E}\left[\sum_{k\in[t]} w(g_k)\right] \tag{1}$$

We claim that the opening costs $w(g_1), \dots, w(g_k)$ are strictly increasing. Towards showing this, fix some $k \in [t-1]$, and suppose that g_k is opened on the arrival of demand x and g_{k+1} on the arrival of demand y. Let F_{k+1} be the set of all open facilities right before g_{k+1} is opened. Define

$$f'_k \coloneqq \operatorname*{argmin}_{f\colon d(f_y^{\mathrm{pred}}, f) \le \frac{1}{2} d(f_y^{\mathrm{pred}}, g_k)} w(f).$$

By definition of the PRED step, and since g_{k+1} was opened during a PRED step, it follows that

$$g_{k+1} = \operatorname*{argmin}_{f\colon d(f_y^{\mathrm{pred}}, f) \le \frac{1}{2} d(f_y^{\mathrm{pred}}, F_{k+1})} w(f).$$

Since $g_k \in F_{k+1}$, clearly $d(f_y^{\mathrm{pred}}, F_{k+1}) \le d(f_y^{\mathrm{pred}}, g_k)$, and so the above two displays imply $w(f'_k) \le w(g_{k+1})$. Moreover, we have that

$$\begin{aligned}
d(f_x^{\mathrm{pred}}, f'_k) &\le d(f'_k, f_y^{\mathrm{pred}}) + d(f_y^{\mathrm{pred}}, f_x^{\mathrm{pred}}) \le \frac{1}{2} d(g_k, f_y^{\mathrm{pred}}) + d(f_y^{\mathrm{pred}}, f_x^{\mathrm{pred}}) \\
&\le \frac{1}{2}\big(d(f_x^{\mathrm{pred}}, g_k) + d(f_x^{\mathrm{pred}}, f_y^{\mathrm{pred}})\big) + d(f_y^{\mathrm{pred}}, f_x^{\mathrm{pred}}) \\
&= \frac{1}{2} d(f_x^{\mathrm{pred}}, g_k) + \frac{3}{2} d(f_y^{\mathrm{pred}}, f_x^{\mathrm{pred}}) \\
&\le \frac{1}{2} d(f_x^{\mathrm{pred}}, g_k) + 3\eta_\infty \\
&< d(f_x^{\mathrm{pred}}, g_k)
\end{aligned}$$

The second inequality is by definition of f'_k. The second-to-last is by $d(f_y^{\mathrm{pred}}, f_x^{\mathrm{pred}}) \le d(f_y^{\mathrm{pred}}, f^*) + d(f^*, f_x^{\mathrm{pred}}) \le 2\eta_\infty$ since both $x, y \in X(f^*)$. The last inequality is because $d(f_x^{\mathrm{pred}}, g_k) \ge d(g_k, f^*) - d(f_x^{\mathrm{pred}}, f^*) > 7\eta_\infty - \eta_\infty = 6\eta_\infty$, since $g_k \in F_b(x_\tau)$ and so $d(f^*, g_k) \ge d(f^*, F_b(x_\tau)) > \min\{7\eta_\infty, 4w(f^*)\} = 7\eta_\infty$ by definition of τ and the assumption in the lemma statement.

Now, consider the moment when the PRED step opens facility g_k. Since $d(f_x^{\mathrm{pred}}, f'_k) < d(f_x^{\mathrm{pred}}, g_k)$ as we established, but the PRED step did not choose to open f'_k, it must be that $w(g_k) < w(f)$. Combining with the fact that $w(f'_k) \le w(g_{k+1})$ which we showed earlier, we obtain that $w(g_k) < w(f) \le w(g_{k+1})$, and we have indeed shown that the opening costs are strictly increasing. Since the opening costs are powers of 2, this directly gives us

$$\sum_{k\in[t]} w(g_k) \le 2w(g_t) \tag{2}$$

To finish the proof, we claim that the opening cost $w(g_t)$ of the last opened facility during the long-distance stage is at most $w(f^*)$. Consider the moment when the PRED step opens facility g_t, and let x be the demand whose arrival caused the opening of g_t. Notice that

$$d(f^*, f_x^{\mathrm{pred}}) \le \eta_\infty < \frac{1}{2} d(f^*, F_b(x)) \le \frac{1}{2} d(f^*, F_b^P(x))$$

where the first step is by definition of η_∞, the second is by definition of τ and the assumption in the lemma, and the third is because $F_b(x) \supseteq F_b^P(x)$. As before, since the PRED step chooses to open g_t instead of f^*, it must be that $w(g_t) \le w(f^*)$. The lemma now follows by combining this last relation with Equations (1) and (2). □

Lemma 4.4. *Suppose that $7\eta_\infty \le 4w(f^*)$. Then it holds that*

$$\mathbb{E}[\mathrm{cost}_M(x_\tau)] \le 3w(f^*) + \frac{3}{2}\sum_{k\in[m]} d(x_k, f^*)$$

Proof. We will bound the opening cost and the connection cost of the MEY step separately. For the opening cost, recalling that the MEY step opens a facility with cost 2^{k-1} with probability $\frac{\delta_{k-1}-\delta_k}{2^k}$ (for δ_k as defined in Algorithm 4.2), we have for any $1 \le k \le m$ that

$$\mathbb{E}\big[w\big(F_n^M(x_k)\big)\cdot \mathbb{1}\big\{w\big(F_n^M(x_k)\big) > w(f^*)\big\}\big] \le \frac{1}{2}\sum_{i>\ell^*}\frac{\delta_{i-1}-\delta_i}{2^i}\cdot 2^i \le \frac{1}{2}\delta_{\ell^*} \le \frac{1}{2}d(x_k, f^*)$$

where in the first step we recalled that $w(f^*) = 2^{\ell^*-1}$, and in the last step we used the fact that, by definition, $\delta_{\ell^*} = \min_{f\in F\cup G_{\ell^*}} d(x_k, f)$, and $f^* \in G_{\ell^*}$. Therefore, since at most one facility is opened in each MEY step, we have

$$\begin{aligned}
\mathbb{E}\big[w\big(F_n^M(x_\tau)\big)\big] &\le \mathbb{E}\left[\max_{f\in F_a^M(x_m)} w(f)\right] \le \mathbb{E}\left[\sum_{f\in F_a^M(x_m)} w(f)\right] \\
&\le w(f^*) + \mathbb{E}\left[\sum_{f\in F_a^M(x_m)} w(f)\cdot \mathbb{1}\{w(f) > w(f^*)\}\right] \\
&\le w(f^*) + \frac{1}{2}\sum_{k\in[m]} d(x_k, f^*)
\end{aligned}$$

and this gives us the bound on the opening cost. For the connection cost, we will show that $d(x_\tau, F_a(x_\tau)) \le 2w(f^*) + d(x_\tau, f^*)$ with probability 1. Let $\delta_0, \ldots, \delta_L$ be defined as in Algorithm 4.2, and recall that $\delta_0 = d(x_\tau, F_b(x_\tau)) \ge d(x_\tau, F_a(x_\tau))$. If $\delta_0 \le 2w(f^*) + d(x_\tau, f^*)$, then the claim holds immediately, so assume otherwise. Let $k' := \max\{k \mid \delta_k > 2w(f^*) + d(x_\tau, f^*)\}$, which exists by assumption, and observe that $k' < \ell^*$ since $\delta_{\ell^*} \le d(x_\tau, f^*)$ by definition of δ_{ℓ^*}. Then, since x_τ is connected to its closest facility in $F_a(x_\tau)$, the event $\{d(x_\tau, F_a(x_\tau)) \le 2w(f^*) + d(x_\tau, f^*)\}$ holds precisely when we open one of the facilities $f_{k'+1}, \ldots, f_L$ at distances $\delta_{k'+1}, \ldots, \delta_L$ (since δ_i is decreasing in i), and so

$$\begin{aligned}
\Pr\big(d\big(x_\tau, F_a^M(x_\tau)\big) \le 2w(f^*) + d(x_\tau, f^*)\big) &= \min\left(\sum_{i=k'+1}^{L}\frac{\delta_{i-1}-\delta_i}{2^i}, 1\right) \\
&\ge \min\left(\sum_{i=k'+1}^{\ell^*}\frac{\delta_{i-1}-\delta_i}{2^{\ell^*}}, 1\right) = \min\left(\frac{\delta_{k'}-\delta_{\ell^*}}{2^{\ell^*}}, 1\right) \\
&\ge \min\left(\frac{2w(f^*) + d(x_\tau, f^*) - \delta_{\ell^*}}{2^{\ell^*}}, 1\right) = 1
\end{aligned}$$

where in the last line we recalled the definition of k' and we used the fact that $w(f^*) = 2^{\ell^*-1}$ and $d(x_\tau, f^*) \ge \delta_{\ell^*}$ by definition of δ_{ℓ^*}. Thus, the connection cost is bounded by $2w(f^*) + d(x_\tau, f^*)$ w.p. 1, and combining with our previous bound for the opening cost, we obtain the lemma. □

Combining Lemmas 4.3 and 4.4, we obtain the following overall bound for the cost in the long-distance stage, for the case where $7\eta_\infty \le 4w(f^*)$.

Corollary 4.5. *Suppose that $7\eta_\infty \le 4w(f^*)$. Then it holds that*

$$\mathbb{E}\left[\sum_{i\le\tau}\mathrm{cost}_M(x_i)\right] \le 5w(f^*) + \frac{3}{2}\sum_{k\in[m]} d(x_k, f^*)$$

For the case where $7\eta_\infty > 4w(f^*)$, the proof of Lemma 4.3 fails, so we need to modify our argument. However, the overall technical details remain similar, as we will now see. We begin with a helper lemma, the proof of which uses a similar argument as the proof of Lemma 4.4.

Lemma 4.6. *Suppose that $7\eta_\infty > 4w(f^*)$. Then, for every $i < \tau$, it holds that $d(x_i, f^*) \ge w(f^*)$*

Proof. Suppose for the sake of contradiction that $d(x_i, f^*) < w(f^*)$ for some $i < \tau$. Let $\delta_0, \dots, \delta_L$ be defined as in Algorithm 4.2. Then the triangle inequality yields

$$\delta_0 = d(x_i, F_b(x_i)) \ge d(f^*, F_b(x_i)) - d(f^*, x_i) \ge d(f^*, F_b(x_i)) - w(f^*) > 3w(f^*)$$

where the second-to-last step is by assumption and the last is because $d(f^*, F_b(x_i)) \ge \min 7\eta_\infty, 4w(f^*) = 4w(f^*)$ by definition of τ and the assumption in the lemma statement. Hence, we can define $k' \coloneqq \min\{k \mid \delta_k \le 3w(f^*)\}$, and we will have $k' \ge 1$.

We claim that, after the arrival of x_i, there exists a facility within a distance of $3w(f^*)$ from x_i. By definition of k', since δ_i is non-increasing in i, a sufficient condition for this to hold is that the MEY step opens some facility among $f_{k'}, \dots, f_L$ (whose distances from x_i are $\delta_{k'}, \dots, \delta_L$). Hence

$$\begin{aligned}
\Pr\big(d(x_i, F_a^M(x_i)) \le 3w(f^*)\big) &\ge \min\left(\sum_{i=k'}^{L} \frac{\delta_{i-1} - \delta_i}{2^i}, 1\right) \\
&\ge \min\left(\sum_{i=k'}^{\ell^*} \frac{\delta_{i-1} - \delta_i}{2^{\ell^*}}, 1\right) = \min\left(\frac{\delta_{k'-1} - \delta_{\ell^*}}{2^{\ell^*}}, 1\right) \\
&\ge \min\left(\frac{3w(f^*) - w(f^*)}{2^{\ell^*}}, 1\right) = 1
\end{aligned}$$

where in the last line we used the definition of k', the fact that $\delta_{\ell^*} \le d(x_i, f^*) < w(f^*)$ by definition of δ_{ℓ^*} and our assumption, and the fact that $w(f^*) = 2^{\ell^*-1}$. Therefore, w.p. 1 it holds that

$$d(f^*, F_b(x_{i+1})) = d\big(f^*, F_a^M(x_i)\big) \le d\big(x_i, F_a^M(x_i)\big) + d(x_i, f^*) \le 4w(f^*)$$

which contradicts the definition of τ. □

We can now bound the total cost in this case, again following a similar analysis as in Lemma 4.4

Lemma 4.7. *Suppose that $7\eta_\infty \le 4w(f^*)$. Then it holds that*

$$\mathbb{E}\left[\sum_{i\le\tau}\mathrm{cost}_M(x_i)\right] \le 4w(f^*) + 6\sum_{k\in[m]} d(x_k, f^*)$$

Proof. Consider the arrival of some demand x_i, and let $\delta_0, \dots, \delta_L$ be defined as in Algorithm 4.2. We separately analyze the connecting and opening cost of x_i. Beginning with the connecting cost, since $F_n^M(x_i) \subseteq F_a^M(x_i)$, we have the bound

$$d\big(x_i, F_a^M(x_i)\big) \le \max\big\{d\big(x_i, F_a^M(x_i) \setminus F_n(x_i)\big), d\big(x_i, F_n^M(x_i)\big)\big\} = \max\big\{\delta_0, d\big(x_i, F_n^M(x_i)\big)\big\} = \delta_0$$

where the last step is because $F_n^M(x_i)$ is either $\emptyset$ or it contains exactly one facility at distance at most δ_0 from x_i. We claim that $d\big(x_i, F_a^M(x_i)\big) \le d(x_i, f^*) + 2w(f^*)$. If $\delta_0 \le d(x_i, f^*) + 2w(f^*)$ then this holds by the above, so assume otherwise. Let $k' \coloneqq \min\{k \,|\, \delta_k \le d(x_i, f^*) + 2w(f^*)\}$, which exists since δ_i is non-increasing

and is at most ℓ^* since $\delta_{\ell^*} \le d(x_i, f^*)$ by definition of ℓ^* and δ_{ℓ^*}. By definition of k', a sufficient condition for our claim to hold is that the MEY step opens some facility among $f_{k'}, \ldots, f_L$ (whose distances from x_i are $\delta_{k'}, \ldots, \delta_L$). Hence

$$\begin{aligned}
\Pr\big(d\big(x_i, F_a^M(x_i)\big) \le d(x_i, f^*) + 2w(f^*)\big) &\ge \min\left(\sum_{i=k'}^{L} \frac{\delta_{i-1} - \delta_i}{2^i}, 1\right) \\
&\ge \min\left(\sum_{i=k'}^{\ell^*} \frac{\delta_{i-1} - \delta_i}{2^{\ell^*}}, 1\right) = \min\left(\frac{\delta_{k'-1} - \delta_{\ell^*}}{2^{\ell^*}}, 1\right) \\
&\ge \min\left(\frac{d(x_i, f^*) + 2w(f^*) - \delta_{\ell^*}}{2^{\ell^*}}, 1\right) = 1
\end{aligned}$$

where in the last line we recalled the definition of k' and we used the fact that $w(f^*) = 2^{\ell^*-1}$ and $d(x_\tau, f^*) \ge \delta_{\ell^*}$ by definition of δ_{ℓ^*}. Therefore, it indeed holds w.p. 1 that $d\big(x_i, F_a^M(x_i)\big) \le d(x_i, f^*) + 2w(f^*)$, and this concludes our bound for the connecting cost.

We now turn to the expected opening cost $\mathbb{E}[w(F_n^M(x_i))]$. Recalling that the MEY step opens a facility with cost 2^{i-1} with probability $\frac{\delta_{i-1}-\delta_i}{2^i}$, we have that

$$\begin{aligned}
\mathbb{E}[w(F_n^M(x_i))] &\le \sum_{i\in[L]} \min\left(\frac{\delta_{i-1} - \delta_i}{2^i}, 1\right) \cdot 2^{i-1} \le \sum_{i \le \ell^*} 2^{i-1} + \sum_{i > \ell^*} \frac{\delta_{i-1} - \delta_i}{2^i} \cdot 2^{i-1} \\
&< 2^{\ell^*} + \frac{1}{2}\delta_{\ell^*} < 2w(f^*) + d(x_i, f^*)
\end{aligned}$$

where the last step is again by definition of ℓ^* and δ_{ℓ^*}.

To conclude the proof, we combine our above bounds with Lemma 4.6, as follows.

$$\begin{aligned}
\mathbb{E}\left[\sum_{i\le\tau} \mathrm{cost}_M(x_i)\right] &= \mathbb{E}\left[\sum_{i\le\tau} \big(d\big(x_i, F_a^M(x_i)\big) + w(F_n^M(x_i))\big)\right] \\
&\le \mathbb{E}\left[\sum_{i\le\tau} (4w(f^*) + 2d(x_i, f^*))\right] \\
&= \mathbb{E}\left[\sum_{i<\tau} (4w(f^*) + 2d(x_i, f^*)) + 2d(x_\tau, f^*)\right] + 4w(f^*) \\
&\le \mathbb{E}\left[6\sum_{i<\tau} d(x_i, f^*) + 2d(x_\tau, f^*)\right] + 4w(f^*) \\
&< \mathbb{E}\left[6\sum_{i\le\tau} d(x_i, f^*)\right] + 4w(f^*) \\
&\le 6\sum_{i\in[m]} d(x_i, f^*) + 4w(f^*)
\end{aligned}$$

□

We can now combine our above results – in particular Corollary 4.5 and Lemma 4.7 – to obtain the following overall bound for the long-distance stage.

Corollary 4.8. *It holds that*

$$\mathbb{E}\left[\sum_{i\le\tau} \mathrm{cost}_M(x_i)\right] \le 5w(f^*) + 6\sum_{k\in[m]} d(x_k, f^*)$$

4.2.2 Short-Distance Stage

We continue our analysis by examining the short-distance stage, i.e. the sub-sequence of demands $\{x_k\}_{k>\tau}$. As we discussed, in this stage, we intuitively expect that some facility near f^* has been opened. To formalize the analysis, we will require some additional notation. First, let ℓ_i be the unique integer such that $2^{\ell_i-1} \leq d(f^*, F_b(x_i)) < 2^{\ell_i}$. Note that ℓ_i can be negative, and if $d(f^*, F_b(x_i)) = 0$ then we set $\ell_i = -\infty$. Since $\{F_b(x_i)\}_{i\in[m]}$ is a non-decreasing sequence of sets, clearly ℓ_i is non-increasing in i. Next, for any integer ℓ, we define $I_\ell \coloneqq \{k > \tau \mid \ell_i = \ell\}$. Recall that $d(f^*, F_b(x_i)) \leq \min\{7\eta_\infty, 4w(f^*)\}$ for all $i > \tau$ by definition of τ, hence, for any ℓ such that $2^{\ell-1} > \min\{7\eta_\infty, 4w(f^*)\}$, we have $I_\ell = \emptyset$. Finally, let $\underline{\ell}$ be the unique integer such that $2^{\underline{\ell}-1} \leq \frac{\mathrm{OPT}}{n} < 2^{\underline{\ell}}$, and let $\bar{\ell}$ be the unique integer such that $2^{\bar{\ell}-1} \leq \min\{7\eta_\infty, 4w(f^*)\} < 2^{\bar{\ell}}$. Given the above definitions, observe that we can partition all demands $\{x_k\}_{k>\tau}$ into groups $I_{\leq\underline{\ell}-1}, I_{\underline{\ell}}, \ldots, I_{\bar{\ell}}$, where $I_{\leq\underline{\ell}-1} \coloneqq \bigcup_{\ell\leq\underline{\ell}-1} I_\ell$; indeed, all demands $\{x_k\}_{k>\tau}$ satisfy $\ell_k \leq \bar{\ell}$ by definition of τ and $\bar{\ell}$.

Having established our notation, we now turn to bounding the total cost of the Mey step during the short-distance stage. We begin by a short argument for the cost of the demands in $I_{\leq\underline{\ell}-1}$.

Lemma 4.9. *It holds that*

$$\mathbb{E}\left[\sum_{k\in I_{\leq\underline{\ell}-1}} \mathrm{cost}_M(x_i)\right] \leq \frac{2|X(f^*)|}{n} \cdot \mathrm{OPT}$$

Proof. By definition of $\underline{\ell}$, for any $k \in I_{\leq\underline{\ell}-1}$, the connection cost of x_k is bounded by

$$d\big(x_k, F_a^M(x_k)\big) \leq d\big(x_k, F_b^M(x_k)\big) \leq 2^{\underline{\ell}-1} \leq \frac{\mathrm{OPT}}{n}.$$

Let $\delta_0, \delta_1, \ldots, \delta_L$ and $f_1, \ldots, f_L$ be defined as in Algorithm 4.2 upon the arrival of x_k. Then, since the Mey step opens a facility with cost 2^{k-1} with probability $\frac{\delta_{k-1}-\delta_k}{2^k}$, we can bound the expected opening cost as

$$\mathbb{E}\big[w\big(F_n^M(x_k)\big)\big] \leq \sum_{k\in[L]} \min\left(\frac{\delta_{k-1}-\delta_k}{2^k}, 1\right) \cdot 2^{k-1} \leq \sum_{k\in[L]} \frac{\delta_{k-1}-\delta_k}{2} \leq \frac{\delta_0}{2} \leq \frac{\mathrm{OPT}}{2n}$$

Summing up our two above displays, we obtain the desired bound as follows

$$\begin{aligned}\mathbb{E}\left[\sum_{k\in I_{\leq\underline{\ell}-1}} \mathrm{cost}_M(x_i)\right] &\leq \mathbb{E}\left[\sum_{k\in I_{\leq\underline{\ell}-1}} \big(d\big(x_k, F_a^M(x_k)\big) + w\big(F_n^M(x_k)\big)\big)\right] \\ &\leq \sum_{k\in I_{\leq\underline{\ell}-1}} \frac{3}{2n} \cdot \mathrm{OPT} < \frac{2|X(f^*)|}{n} \cdot \mathrm{OPT}\end{aligned}$$

□

To bound the cost for the rest of the demands, we will require a more involved argument, which we now detail.

Lemma 4.10. *For any $\ell \in [\underline{\ell}, \bar{\ell}]$, it holds that*

$$\mathbb{E}\left[\sum_{k\in I_\ell} \mathrm{cost}_M(x_i)\right] \leq 18 \sum_{k\in I_\ell} d(x_i, f^*) + 32w(f^*)$$

Proof. In all the above proofs, we have restricted our attention to demands in $X(f^*)$, and we have relabeled their indices so that $X(f^*) = \{x_1, \ldots, x_m\}$. In this proof, we will also need to argue about demands outside of $X(f^*)$. Hence, we will denote $X = \{y_1, \ldots, y_n\}$ the complete set of demands in order of arrival, and we define $\sigma\colon [m] \to [n]$ the relabeling that maps each element $x_j \in X(f^*)$ to its identity $y_i \in X$, i.e. $\sigma(j) = i$.

For integer ℓ, define $\tau_\ell \coloneqq \min\{i \in [n] \mid F_n^M(y_i) \neq \emptyset \quad \text{and} \quad d(f^*, F_n^M(y_i)) < 2^{\ell-1}\}$. By definition of I_ℓ, we have that

$$\mathbb{E}\left[\sum_{i \in I_\ell} \text{cost}_M(x_i) - 18d(x_i, f^*)\right] = \mathbb{E}\left[\sum_{i \in I_\ell} (\text{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\}\right]$$

Hence, to prove the lemma, it suffices to show that

$$\mathbb{E}\left[\sum_{i \in [\tau_\ell]} (\text{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\}\right] \leq 32w(f^*)$$

We will show this via an inductive argument. Specifically, let $\mathcal{F}_j \coloneqq (F_n^M(x_1), F_n^P(x_1), \ldots, F_n^M(x_j), F_n^P(x_j))$. We will show that, for any $j \in [n]$, it holds that

$$\mathbb{E}\left[\sum_{i=j}^{\tau_\ell} (\text{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_{j-1}\right] \leq 32w(f^*) \tag{3}$$

Clearly, the desired bound holds by applying Equation (3) for $j = 1$, so it suffices to prove this last statement. We will do so by backward induction on j. We begin by establishing the base case $j = n$.

Base case. Note that, for any $j \in [n]$, having conditioned on $\mathcal{F}_{j-1}$, the variables $\delta_0, \ldots, \delta_L$ defined on the arrival of y_j are deterministic, and so is the event $\{j \in \sigma(I_\ell)\}$. Thus, recalling again that the MEY step opens a facility with cost 2^{i-1} with probability $\frac{\delta_{i-1} - \delta_i}{2^i}$, we have

$$\mathbb{E}[\text{cost}_M(y_j) \mid \mathcal{F}_{j-1}] \leq d(y_j, F_b(y_j)) + \sum_{i \in [L]} \frac{\delta_{i-1} - \delta_i}{2^i} \cdot 2^{i-1} < 2d(y_j, F_b(y_j)) \tag{4}$$

since $\delta_L \leq d(y_j, F_b(y_j))$ by definition of δ_L. Therefore, we obtain

$$\begin{aligned}
\mathbb{E}\left[\sum_{i=n}^{\tau_\ell} (\text{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_{j-1}\right] &\leq \mathbb{E}[(\text{cost}_M(y_n) - 18d(y_n, f^*)) \cdot \mathbb{1}\{n \in \sigma(I_\ell)\} \mid \mathcal{F}_{n-1}] \\
&\leq 2d(y_n, F_b(y_n)) - 18d(y_n, f^*)|_{n \in \sigma(I_\ell)} \\
&\leq 2d(y_n, F_b(y_n)) - 2d(y_n, f^*)|_{n \in \sigma(I_\ell)} \\
&\leq 2d(f^*, F_b(y_n))|_{n \in \sigma(I_\ell)} \\
&\leq 2^{\ell+1} \leq 2^{\ell^*+1} = 4w(f^*)
\end{aligned}$$

where the second line is by Equation (4), the second-to-last is by the triangle inequality, and the last is by definition of I_ℓ. Hence, Equation (3) follows.

Inductive step. Having shown the base case for $j = n$, we now assume that Equation (3) holds for $j+1, j+2, \ldots, n$ and we prove that it also holds for j. We discern the following three cases.

Case 1: $j \notin \sigma(I_\ell)$.

By the law of total expectation, we have

$$\begin{aligned}
&\mathbb{E}\left[\sum_{i=j}^{\tau_\ell} (\text{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_{j-1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\sum_{i=j+1}^{\tau_\ell} (\text{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_j\right] \,\middle|\, \mathcal{F}_{j-1}\right] \\
&\leq 32w(f^*)
\end{aligned}$$

where the second step is because $\mathbb{1}\{j \in \sigma(I_\ell)\} = 0$, and the last step follows by the induction hypothesis.

Case 2: $j \in I_\ell$ and $8d(y_j, f^*) > d(f^*, F_b(y_j))$

In this case, we have

$$d(y_j, F_a(y_j)) \le d(y_j, f^*) + d(f^*, F_b(y_j)) \le 9d(y_j, f^*)$$

Therefore, Equation (4) implies

$$\mathbb{E}[\mathrm{cost}_M(y_j) - 18d(y_j, f^*) \mid \mathcal{F}_{j-1}] \le 2d(y_j, F_b(y_j)) - 18d(y_j, f^*) \le 0$$

and Equation (3) now follows similarly in Case 1, that is

$$\begin{aligned}
&\mathbb{E}\left[\sum_{i=j}^{\tau_\ell}(\mathrm{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_{j-1}\right] \\
&\le \mathbb{E}\left[\mathbb{E}\left[\sum_{i=j+1}^{\tau_\ell}(\mathrm{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_j\right] \,\middle|\, \mathcal{F}_{j-1}\right] \\
&\le 32w(f^*)
\end{aligned}$$

Case 3: $j \in I_\ell$ and $8d(y_j, f^*) \le d(f^*, F_b(y_j))$

In this case, we have

$$\delta_{\ell^*} \le d(y_j, f^*) \le \frac{1}{8}d(f^*, F_b(Y_j)) \le 2^{\ell-3}$$

Let $k' \coloneqq \min\{k \mid \delta_k \le 2^{\ell-2}\}$. Then, for all $i \ge k'$, it holds that

$$d(f_i, f^*) \le d(y_j, f_i) + d(y_j, f^*) < 2^{\ell-1}$$

Also, let $D \coloneqq \{f \colon d(f^*, f) \ge 2^{\ell-1}\}$, and let $\mathcal{E}$ be the event that, on the arrival of y_j, the MEY step either opens a facility in D or does not open a facility. Recalling the definition of τ_ℓ, it is immediate that

$$\begin{aligned}
\Pr(\tau_\ell \le j \mid \mathcal{F}_{j-1}) &\ge \Pr(\mathcal{E} \mid \mathcal{F}_{j-1}) \\
&\ge \Pr(f_i \text{ is open for some } i \ge k \mid \mathcal{F}_{j-1}) = \min\left(\sum_{i \ge k}\frac{\delta_{i-1} - \delta_i}{2^i}, 1\right) \\
&\ge \min\left(\frac{\delta_{k-1} - \delta_{\ell^*}}{2^{\ell^*}}, 1\right) \ge \min\left(\frac{2^{\ell-2} - 2^{\ell-3}}{2^{\ell^*}}, 1\right) \ge \min\left(2^{\ell-3-\ell^*}, 1\right) \\
&\ge \frac{\delta_0}{16w(f^*)} \ge \frac{\mathbb{E}[\mathrm{cost}_M(y_j) \mid \mathcal{F}_{j-1}]}{32w(f^*)}
\end{aligned}$$

where the second to last inequality follows from $\delta_0 \le 2^\ell$ by definition of δ_0 and the fact that $j \in I_\ell$, as well as the fact that $w(f^*) = 2^{\ell^*-1}$. Therefore, we obtain that

$$\begin{aligned}
&\mathbb{E}\left[\sum_{i=j}^{\tau_\ell}(\mathrm{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_{j-1}\right] \\
&= \mathbb{E}[\mathrm{cost}_M(y_j) \mid \mathcal{F}_{j-1}] + \mathbb{E}\left[\mathbb{E}\left[\sum_{i=j+1}^{\tau_\ell}(\mathrm{cost}_M(y_i) - 18d(y_i, f^*)) \cdot \mathbb{1}\{i \in \sigma(I_\ell)\} \,\middle|\, \mathcal{F}_j, \mathcal{E}\right] \,\middle|\, \mathcal{F}_{j-1}\right] \\
&\le \Pr(\bar{\mathcal{E}} \mid \mathcal{F}_{j-1}) \cdot 32w(f^*) + \Pr(\mathcal{E} \mid \mathcal{F}_{j-1}) \cdot 32w(f^*) \\
&= 32w(f^*)
\end{aligned}$$

where the second step follows by the inductive hypothesis. Hence, we have again shown Equation (3). □

4.2.3 Putting everything together

We are now ready to combine our bounds for the cost in the long- and short-distance stages, in order to prove Lemma 4.2. Indeed, using Lemmas 4.3, 4.4, 4.6 and 4.7, we have that

$$\begin{aligned}
\sum_{i\in[m]} \mathbb{E}[\text{cost}_M(x_i)] &= \mathbb{E}\left[\sum_{i<\tau} \text{cost}_M(x_\tau)\right] + \mathbb{E}[\text{cost}_M(x_\tau)] + \sum_{\ell=\underline{\ell}}^{\bar{\ell}} \mathbb{E}\left[\sum_{i\in I_\ell} \text{cost}_M(x_i)\right] + \sum_{i\in I_{\leq \underline{\ell}-1}} \mathbb{E}[\text{cost}_M(x_\tau)] \\
&\leq 2w(f^*) + \left(2w(f^*) + 2\sum_{i\in[m]} d(x_i, f^*)\right) + \sum_{\ell=\underline{\ell}}^{\bar{\ell}} \left(18\sum_{i\in I_\ell} d(x_i, f^*) + 32w(f^*)\right) \\
&\qquad + \frac{2|X(f^*)|}{n} \cdot \text{OPT} \\
&\leq \big(O * (\bar{\ell} - \underline{\ell}) + O(1)\big) \cdot \left(\sum_{i\in[m]} d(x_i, f^*) + w(f^*)\right) + \frac{2|X(f^*)|}{n} \cdot \text{OPT} \\
&\leq O\left(\max\left\{1, \log \frac{n\eta_\infty}{\text{OPT}}\right\}\right) \cdot \left(\sum_{i\in[m]} d(x_i, f^*) + w(f^*)\right) + \frac{2|X(f^*)|}{n} \cdot \text{OPT}
\end{aligned}$$

5 Online Bipartite Matching

We'll refocus our discussion with *Online Bipartite Matching*. Continuing away from LPs, the offline version of this problem is fairly simple and largely well known, solvable optimally in polynomial time. The offline version involves a graph $G = (U \cup V, E)$ where E is guaranteed to only make connections between vertices in U with vertices in V. In this problem we seek to match as many vertices as possible wherein a match occurs between exactly two vertices connected across an edge and a vertex can only be involved in at most one match. Actually counted will be the number of matched pairs, though it must be clear that only the size of this matching is important; the exact pairs are not significant. The online version separates distinctly U and V, providing U and $|V| = n$ in advance and slowly revealing the vertices of V one-at-a-time in addition to their neighbors across E, $N(v) \subseteq U, v \in V$, and requiring an irrevocable decision at that time of which neighboring vertex in U (or no vertex) with which v will be matched[KVV90]. Our goal remains to, in the online course of the problem, seek to maximize this number of matches.

Introduced in [KVV90] in part as an exploration of the power of randomness in online algorithms, they were able to find that a competitive ratio of $1 - \frac{1}{e}$ was achievable through an overall greedy approach, but that a ratio above $1/2$ was not possible for deterministic algorithms in an adversarial setting. The problem has since been discussed in a variety of attempts to improve the worst case, but without significant digression from greedy algorithms as discussed experimentally in [BKP18]. However, in [CI21] and [CGLB24] the problem is expanded to consider learning-augmented versions of the problem which permit some additional possibility of leveraging potentially faulty advice for the problem.

Namely, [CGLB24] considers a version of the problem partially leveraging an arbitrary underlining advice-free algorithm to the online bipartite matching problem with additional advice based neighbor information about the graph. Also discussed in [BKP18], type information encompasses the set of (particular) neighbors are held by online vertices in the problem. A summary of type counts across the problem will form the basis of predictions and subsequent advice relevant to this algorithm. These type counts are well construed as the number of times each subset of U is connected to by some vertex $v \in V$.

It is critical to limit the scope of this discussion somewhat. With an argument of fundamental nature to the original [KVV90] argument against the plausibility of solutions better than $1/2$ in the simple adversarial ordering version, adversarial arrival remains to difficult a context for learning-augmented algorithms and the paper [CGLB24] provides an argument establishing that learning-augmentation is not meaningfully helpful. Instead, we'll limit our consideration to the random ordering version of the problem.

Algorithm 5.1: Learning-Augmented Online Bipartite Matching

Input: Online graph $G = (U \cup V, E)$, randomized arrival ordering v_i, $1 \leq i \leq n$, $n = |V|$; advice type vector $\hat{c}$ with $\hat{r}$ nonzero entries, advice-free algorithm BASELINE which has competitive ratio $\beta < 1$, error threshold $\varepsilon > 0$, failure rate $\delta = \delta' + \delta_{poi}$ for $\delta_{poi} = O(\frac{1}{poly(\hat{r})})$

Procedure TESTANDMATCH():
  // Where $\hat{n}$ is the optimal advice matching count
  Compute optimal advice matching $\hat{M}$ from $\hat{c}$
  if $\frac{\hat{n}}{n} \leq \beta$ **then**
    Run BASELINE on all
  $s_{\hat{r},\varepsilon,\delta} \leftarrow O(\frac{(\hat{r}+1)\log(1/\delta')}{\varepsilon^2 \log(\hat{r}+1)})$
  $\tau \leftarrow 2(\frac{\hat{n}}{n} - \beta) - \varepsilon$
  Run MIMIC on $s_{\hat{r},\varepsilon,\delta} \cdot \sqrt{\log(\hat{r}+1)}$ arrivals, keep track of the online arrivals in a set A
  if MINIMAXTEST $(s_{\hat{r},\varepsilon,\delta}, \frac{\hat{c}}{n}, A, \tau, \delta')$ **then**
    Run MIMIC on the remaining arrivals
  else
    Run BASELINE on the remaining arrivals

Procedure MIMIC *(matching $\hat{M}$, advice counts $\hat{m}$, arrival type label t)*:
  if $\hat{m}[t] > 0$ **then**
    Mimic arbitrary unused type t match in $\hat{M}$
    Decrement $\hat{m}[t]$ by 1
  return $\hat{m}$

This portion, Algorithm 5.1, is first submitted by [CGLB24] and analyzed therein. We should gather an intuition by first overviewing the behaviour of the algorithm. Given a set of types and without care to the identity of particular vertices, there is a unique construction *shape* of the matching graph. Since we now have a definite graph, we can leverage offline algorithms to achieve an optimal matching on the advice graph, yielding an advice matching. This is an optimal solution, assuming the type advice in $\hat{c}$ is actually correct. There is merited some brief discussion that this can be done in efficiently overall as, although $\hat{c}$ has $2^{|U|}$ entries, at most $|V|$ will have nonzero values. This warrants a brief discussion as to efficiency as, if this data is maintained in a sparse format, L_1 distance between two type containers need only consider the interaction between one container and the other then the sum of non-interacting which is calculable from sum information only. Moreover, graph construction need only consider these nonzero components as well. We can guarantee that at worst all online $v \in V$ have their own type and thus cause a nonzero value, so the data and its usage remain efficient (though, without considering efficiency in the creation of this advice). If the advice matching fails to do better than the worst-case of the provided BASELINE, merely use BASELINE.

Then we define a value s relating to sample size needed which is dependent on both the size of the advice and parameters regarding the behaviour of the algorithm at large. Also defines is a testing threshold primarily in use for the analysis later. We will discuss the nature of MINIMAXTEST in a moment, but for now it is sufficient to conceive of the test as a 'closeness' between our observed online arrivals so far and the received advice. If we are sufficiently confident that our advice is 'close enough' to the real distribution, we continue with MIMIC and achieve the portion of the matching which does align with the real distribution. If, however, we fail to appear to be 'close enough', revert to running BASELINE which will achieve β across the remaining arrivals, though notably with worse performance guarantees overall than just using BASELINE, because TESTANDMATCH already used arrivals under MIMIC.

The portions in Algorithm 5.2 primarily originate instead from [JHW18] with modification in [CGLB24]. Here the MINIMAXTEST relies on a series of proofs which submit that with high probability we can expect the L_1 distance between a distribution yielding online arrivals A and another distribution, for us $\hat{c}/n$, to be sufficiently low that it is within our error threshold.

Algorithm 5.2: IID from Online

Input: An appropriate polynomial-time L_1 estimation algorithm J

Procedure MinimaxTest *($s, \frac{\hat{c}}{n}, A, \tau, \delta'$)*:
  Sample $s_1, s_2 \sim \text{Poisson}(s/2)$
  if $s_1 + s_2 > s\sqrt{\log(\hat{r}+1)}$ **then**
    return *False*
  Collect $s_1 + s_2$ IID samples from $p^* = \frac{c^*}{n}$ via SimulateP (A)
  Run an algorithm J to estimate $\hat{L}_1$ such that $|\hat{L}_1 - L_1(p,q)| \leq \varepsilon$ w.p. $1 - \delta'$
  return $\hat{L}_1 < \tau$

Procedure SimulateP *(Array A, desired samples $s \leq |A|$)*:
  $T^s_{p^*} \leftarrow \emptyset$
  $i \leftarrow 0$
  while $|T^s_{p^*}| < s$ **do**
    if *Bernoulli*$(i/n) = 1$ **then**
      $x \leftarrow$ uniform random selection from $\{A[0], ..., A[i-1]\}$
    else
      $x \leftarrow A[i]$
      $i \leftarrow i + 1$
    $T^s_{p^*} \leftarrow T^s_{p^*} + x$
  return $T^s_{p^*}$

There also exists an apparent hole in MinimaxTest in that no specific algorithm is specified for this estimation. We address this also with [JHW18], which offers of a method for the determination of an algorithm to satisfy these probabilistic and error requirements which is guaranteed to exist for our definition methods of s, $\hat{r}$, and for $\varepsilon > 0$, $1 - \delta < 1$. Thus although this explanation will consider the mathematical proof of this algorithm's nature to be outside of the scope of this report, we may rest assured that it exists and is useful at least to the requirements placed upon it in this analysis. We will more exactly take by this construction that, from [JHW18], both Lemma 5.1 and Lemma 5.2:

Lemma 5.1. *For fixed reference distribution q with domain T, $|T| = r$ and where $s \in O(\frac{r \log(1/\delta)}{\varepsilon^2 \log r})$ is an even integer, then there exists a constructable algorithm which draws $s_1 + s_2$ IID samples from unknown distribution p over T. This with $s_1, s_2 \sim$ Poisson$(s/2)$. That algorithm yields an estimate $\hat{L}_1$ wherein $|\hat{L}_1 - L_1(p,q)| \leq \varepsilon$ w.p. at least $1 - \delta$.*

Lemma 5.2. *For any $m > 0$ and any $x > 0$, then where $X \sim$ Poisson(m), $Pr[|X - m| \geq x] \leq 2e^{-\frac{x^2}{2(m+x)}}$*

Now in the analysis of the TestAndMatch algorithm itself, we'll seek to achieve a 1-consistent algorithm (from perfect advice), yet a near-β-robust algorithm for non-augmented β-CR algorithm (with arbitrarily bad advice).

Since $\hat{M}$ is unique to $\hat{c}$ and since type information, as discussed, well describes the graph, and since solving $\hat{M}$ creates an optimal matching we can get $\hat{n}$ as the best matching count for the advice. We form $\hat{m}$ used in Mimic as the number of times $\hat{M}$ uses each type as being matched in a particular form. Thus, Mimic, while it has the capacity to obey $\hat{M}$, continues to dedicate entries to this matching in accordance to that capacity. If, instead, there exists no capacity apparently remaining in $\hat{M}$, Mimic merely discards the match. Therefore, fully using Mimic only tries to get as close to $\hat{M}$ as possible and does not make greedy selections beyond that.

Lemma 5.3. *Let $p^* = c^*/n$ and $q = \hat{c}/n$ for offline type count c^*.* Mimic *outperforms* Baseline *in the worst case if and only if $\frac{\hat{n} - L_1(c^*, \hat{c})/2}{n^*} > \beta \iff L_1(p^*, q) < \frac{2}{n}(\hat{n} - \beta n^*)$*

Proof. Let $\hat{m}_t$ be the matching count for a particular type t in $\hat{M}$, note the sum of these must also be

the advice matching success $\hat{n}$. MIMIC in application achieves $m = \sum_t \min(c_t^*, \hat{m}_t)$ as at least that many matching points will exist and only m are actually matched in $\hat{M}$.

Further $m = \sum_t \min(c_t^*, \hat{m}_t) = \sum_{t:c_t^* \geq \hat{m}_t} \hat{m}_t + \sum_{t:c_t^8 < \hat{m}_t} c_t^* = \hat{n} - \sum_{t:c_t^* < \hat{m}_t} (\hat{m}_t - c_t^*)$.

Identify that as $\hat{m}_t \leq \hat{c}_t$ for all t as a type must be expected to be a match. As $\min(a,b) = \frac{a+b-|a-b|}{2}$ and as $\sum_t c_t^* = \sum_t \hat{c}_t = n$, we acknowledge that $L_1(a,b) = 2\sum_{t:a_t<b_t}(b,a)$. Thus, $m = \hat{n} - \frac{1}{2}L_1(c^*, \hat{m}) \leq \hat{n} - \frac{1}{2}L_1(c^*, \hat{c})$

So blindly following MIMIC advice, the algorithm achieves at least $\hat{n} - L_1(c^*, \hat{c})/2$. Thus we get a competitive ratio of $\frac{\hat{n} - L_1(c^*,\hat{c})/2}{n^*}$. This is clearly only better when it exceeds β from BASELINE, which occurs when, by algebraic reordering, $L_1(c^*, \hat{c}) < 2(\hat{n} - \beta n^*)$. □

To make stronger guarantees without knowing the optimal match count n^* in advance, merely use n instead as this makes the check more conservative and increases 'closeness' demands to use MIMIC only. In doing so, even without knowing the true value, we can still guarantee a better-than-β competitive ratio for MIMIC. Since Algorithm 5.2's MINIMAXTEST does make this guarantee in order to use MIMIC on the second portion of the algorithm, then should the MINIMAXTEST pass, we have used MIMIC the entire time and get this better-than-β guarantee. So we guarantee we have information to make this claim at high probability following Lemma 5.1. This demands a tight domain, the aforementioned scarcity limits the relevant domain to merely $\hat{r} + 1 \leq n + 1$. Then by providing following the MINIMAXTEST scheme and leveraging the lemma we get a high-probability estimation that the L_1 distance is within our error and scale tolerance. If instead the number of samples present in A is insufficient, MINIMAXTEST returns false and we return to BASELINE which we will analyze more of later.

Lemma 5.4. MINIMAXTEST *passes with probability at least* $1 - \delta'$ *when there are sufficient online arrivals in* A *and that thus* $L_1(p^*, q) < 2(\frac{\hat{n}}{n} - \beta)$

Proof. Since we have the precondition of sufficient online arrivals, MINIMAXTEST passes the first condition.

Following Lemma 5.1, our estimate $\hat{L}_1$ is underestimated by at most ε w.p. at least $1 - \delta'$.

Since to pass $\hat{L}_1 < \tau$, at worst $L_1^* - \varepsilon < 2(\frac{\hat{n}}{n} - \beta) - \epsilon$ with the same probability. □

Let us take a moment to identify when there are insufficient arrivals, however. Since we will actually collect $s\sqrt{\log(\hat{r}+1)}$, failure is primarily dependent on the sum of s_1, s_2 exceeding by chance this value. Since we can sum Poisson probabilities and check the size of the tail, we find the probability of failure due to this $\delta_{poi} = Pr[|(s_1 + s_2 - s)| > s\sqrt{\log \hat{r}}]$ which by Lemma 5.2 is at most $2e^{-\frac{s^2 \log \hat{r}}{2(s + s\sqrt{\log \hat{r}})}}$. For most of the analysis, we will simplify this to $O(\frac{1}{poly(\hat{r})})$ given the value of s selected in TESTANDMATCH.

So when TESTANDMATCH does eventually choose to run MIMIC, we can be highly confident that the result is good. However, this does not yet address the possible case that MINIMAXTEST fails and we return to BASELINE.

Lemma 5.5. *Beginning with arbitrary algorithm for the first* $k \in [n]$ *online arrivals which makes* j *matches, subsequently switching to* BASELINE*, the overall matching will be at least* $\beta(n - k - j) + j$

Proof. First we identify that for even an arbitrary algorithm, any particular match reduces the total possible number of matches by at most 2 including both the consumption of the online vertex and maximum possible blocking from the use of some offline vertex.

Thus, the remaining possible number of matches following BASELINE is guaranteed to be at least $\beta(n-k-j)$ as $n - k$ online vertices remain which could be matched and we reduce this by another j for the *additional* maximum possible number of offline vertices disqualified from a matching. We therefore run BASELINE on the remaining $n - k - j$ and are guaranteed at least rate β on that quantity.

Since j matches have occurred already, we do slightly better with an additional j matches from the first section. □

Since, of course, MIMIC is within the category of arbitrary algorithms, we know our solution must do at least this well.

Theorem 5.6. TESTANDMATCH *for the constraints of the inputs achieves a competitive ratio of at least* $\frac{\hat{n}}{n} - \frac{L_1(p,q)}{2} \geq \beta$ *with good advice* $\hat{L}_1 \leq 2(\frac{\hat{n}}{n} - \beta) - \varepsilon$ *and at least* $\beta(1 - \frac{k}{n})$ *otherwise with success probability at least* $1 - \delta$*, where* $\delta = \delta' + \delta_{poi}$*.*

Proof. Let us begin with consideration of the two basic cases.

In the 'good' case, $\hat{L}_1 \leq 2(\frac{\hat{n}}{n} - \beta) - \varepsilon$. Following the discussion immediately after Lemma 5.4, we are $\delta_{poi} \leq O(\frac{1}{poly(\hat{r})})$ likely to have sufficient arrivals after TESTANDMATCH gathers $s_{\hat{r},\varepsilon,\delta} \cdot \sqrt{\log(\hat{r}+1)}$. Then, using Lemma 5.4, we are $1 - \delta'$ likely to find that $L_1(p^*, q) < 2(\frac{\hat{n}}{n} - \beta)$. Thus we have accumulated error at worst $\delta = \delta' + \delta_{poi}$. Then following Lemma 5.3, we find that we get a matching m, $\frac{m}{n^*} \geq \frac{m}{n} \geq \frac{\hat{n} - L_1(c^*, \hat{c})/2}{n} \geq \beta$.

In the 'bad' case $\hat{L}_1 > 2(\frac{\hat{n}}{n} - \beta) - \varepsilon$. After $k = s_{\hat{r},\varepsilon,\delta} \cdot \sqrt{\log(\hat{r}+1)}$ arrivals MINIMAXTEST fails, so we return to BASELINE and follow Lemma 5.5. Consequently, we achieve at worst $\frac{m}{n^*} \geq \frac{m}{n} \geq \beta \cdot (1 - \frac{k}{n}) = \beta \cdot (1 - \frac{s_{\hat{r},\varepsilon,\delta} \cdot \sqrt{\log(\hat{r}+1)}}{n})$ as the competitive ratio. However, as we have defined $s_{\hat{r},\varepsilon,\delta}$ appropriately, we can find this to be an $o(1)$ cost. □

Although for perfect advice this places a high probability that we achieve a consistency of 1, we note that the robustness is only *near* β. Indeed, robustness is slightly worse than merely using the best known non-augmented algorithm at β. Moreover, this loss is not strongly constant, following $\frac{(\hat{r}+1)\log(1/\delta')}{\varepsilon^2 n \sqrt{\log(\hat{r}+1)}}$. However, considering that this value tends to be small when $\hat{r} < n$, which is plausible, and considering the plausibility of a better score is high for a reasonably good predictor, there is meaningful value to considering this approach.

Thus for all instances and any advice, we can conclude the algorithm achieves competitive ratio of at least $\beta(1 - o(1))$.

Moreover this framework merits some additional credit if only for its relative anonymity to underlying algorithms. Even without active reconsideration of advice or strong integration of predictions, the algorithm achieves a strong result analogous to that of the previous problems. This type of 'switcher algorithm' is fundamentally more dynamic and allows for growth as open problem are solved, while also requiring less problem-wise consideration than strong integration may require.

6 Metrical Task Systems

Metrical Task Systems is a markedly general online problem meaningfully related to applications in task scheduling and in caching, which can be formulated as a special case. Originally introduced in [BLS92], the problem has since been learning-augmented in [ACEPS23]. As with other learning-augmented variations, we will seek to achieve both good consistency and a robustness comparable to the best known non-augmented online algorithms.

In the problem we are given a metric space M which contains states S and has a metric d cost between these states. Starting at an initial state x_0, we receive online a cost function ℓ_t over M to $\mathbb{R}_{\geq 0} \cup \{\infty\}$ and we must decide to move to another state (or not move) resulting in state x_t and incur total cost $d(x_{t-1}, x_t) + \ell_t(x_t)$. The objective is to minimize the total cost $\sum_{t=1}^{T}(d(x_{t-1}, x_t) + \ell_t(x_t))$ [BLS92]. For the learning-augmentation, we also receive from a predictor a prediction with error η_t which is the state distance between the predictor and some, potentially optimal, offline algorithm. We take the sum of all errors to be the prediction error η [ACEPS23].

It's critical to note that blindly obeying even a somewhat accurate, but imperfect, predictor can incur irrecoverable costs due to the fact that ℓ_t could be ∞ for a particular state, regardless of if a nearby state has a much better cost. Should our algorithm take such decisions blindly, robustness will collapse rapidly. It is also valuable to identify that η on its own is a difficult to comprehend figure, being partially dependent on the actual size of the metric space, so instead we will ground it through the cost of the offline algorithm

through η/c_{OFF}. We will continue to use this $c_{\text{algorithm}}$ notation to identify the cost of an algorithm. This, assuming its optimal is above 0, does provide some information as to the relativity of the error to the particular problem instance.

Algorithm 6.1: Follow the Prediction

Input: Given time step t, which carries cost function ℓ_t, and a prediction therefor p_t

Procedure `FTP`(t, ℓ_t, p_t)**:**

  $x_t \leftarrow \text{argmin}_{x \in X}(\ell_t(x) + 2d(x, p_t))$

Thus Algorithm 6.1, presented in [ACEPS23], merely follows the prediction so long as it is not better to move to another state, pay its cost, and move back to the predicted state.

Lemma 6.1. *For any M with predictions, Algorithm 6.1 achieves a competitive ratio of $1 + \frac{4\eta}{c_{OFF}}$, against OFF with η for predictions relative to OFF.*

Proof. Let A_t be an algorithm which agrees with `FTP` from states x_0 and predicted states p_1 through p_t but then follows an offline algorithm's states o_{t+1} through o_T. Consider two instances of the algorithm A_t and A_{t-1}.

By the definition of the algorithm, the worst cost incurred by an algorithm in any particular step step is $\ell_t(x_t) + 2d(x_t, x_{t-1})$. Since both algorithms follow the same path with exception to the step t, we need only consider this step where x_t may not be o_t. So $c_{A_t} - c_{A_{t-1}} = d(x_{t-1}, x_t) - d(x_{t-1}, o_t) + \ell_t(x_t) - \ell_t(o_t) + d(x_t, o_{t+1}) - d(o_t, o_{t+1})$. Then, by the triangle inequality, $c_{A_t} - c_{A_{t-1}} \leq d(x_t, o_t) + \ell_t(x_t) - \ell_t(o_t) + d(x_t, o_t)$, thus $c_{A_t} \leq c_{A_{t-1}} + \ell_t(x_t) - \ell_t(o_t) + 2d(x_t, o_t)$. Similarly, $c_{A_t} \leq c_{A_{t-1}} + 2d(o_t, p_t) - \ell_t(o_t) + \ell_t(x_t) + 2d(p_t, x_t)$

So then taking from the argmin in Algorithm 6.1, $\ell_t(x_t) + 2d(p_t, x_t) \leq \ell_t(o_t) + 2d(p_t, o_t)$, $c_{A_t} \leq c_{A_{t-1}} + 4d(o_t, p_t)$ as the x_t selected is no worse than the optimal in that argmin for that step.

Thus, as the running on the algorithm on all points achieves exactly c_{A_T} and the offline algorithm is exactly A_0, by sum, $c_{\text{FTP}} = c_{A_T} \leq c_{A_0} + 4\sum_{t=1}^{T} \eta_t = c_{\text{OFF}} + 4\eta$ □

So now we can make some guarantees regardless of prediction quality, however this is not sufficient to make a robustness guarantee as η, the error, directly worsens the competitive ratio. We need to seek some limitation on the growth of this 'badness'. Thus, we consider layered schemes to achieve a robustness guarantee.

The deterministic case achieves a strict robustness bound, leveraging Algorithm 6.2 from [FRR94].

Algorithm 6.2: MinDet

Input: A parameter $1 < \gamma \leq 2$, set of online algorithms A, input sequence I

Procedure `MinDet()`**:**

  $l \leftarrow 0$

  Start tracking all $A_i \in A$

  while *inputs remain* **do**

    $i \leftarrow l \mod m$

    if $c_{A_i} \leq \gamma^l$ **then**

      follow A_i

      track all algorithm steps

    $l \leftarrow l + 1$

In an effort to generalize the result from [FRR94], [ACEPS23] performs the following generalizing argument shown in Lemma 6.2. Notably, we will disqualify all zero-cost problems, which are largely disinteresting in an MTS context regardless. Any cost $c \in (0, 1)$ can be moved to 1 by scaling the problem.

Lemma 6.2. *Assume $c_{OPT_I} > 0$. Given the set of A online algorithms for a problem P in MTS with $OPT_I \geq 1$,* `MinDet` *achieves at most $(2\frac{\gamma^{|A|}}{\gamma - 1} + 1) \cdot \min_i(c_{A_i}(I))$, for any input sequence I.*

Proof. On cycle l of the algorithm, we note $i = l \mod m$ and let $i' = (l-1) \mod m$. Consider switching from algorithm $A_{i'}$ with current cost $c'_{A_{i'}} = \gamma^{l-1}$ to A_i with end-of-cycle cost c_{A_i}.

Cost during the cycle, for both switching and execution, is at most $c'_{A_{i'}} + c'_{A_i} + (c_{A_i} - c'_{A_i}) = c'_{A_{i'}} + c_{A_i} = \gamma^{l-1} + \gamma^l$ as, at worst, you must take the entire cost of $A_{i'}$ up to where we leave it, but can reach the new state from the new algorithm, at worst, by following the entire metric distance behaviour of A_i which is at worst c_{A_i}.

Therefore, landing at the last cycle L using algorithm $i = L \mod m$ we can bound MinDet at $c_{\text{MinDet}} \leq 2\sum_{l=1}^{L-1} \gamma^l + c_{A_i} \leq 2\frac{\gamma^L}{\gamma-1} + c_{A_i}$ by geometric sum and an inequality.

If $L < m$ such that we have not repeated any algorithm, we leverage OPT ≥ 1 so $c_{A_i} \geq 1 \; \forall A_i \in A$. Note that $L + 1 \leq m$ Thus, we can make the guarantee that $c_{\text{MinDet}} \leq 2\frac{\gamma^{L+1}}{\gamma-1} \leq 2\frac{\gamma^m}{\gamma-1} \cdot \min_i(c_{A_i})$.

If instead $L \geq m$, we must have repeated at least one algorithm. We will first consider if, with $i = L \mod m$ and where A_i has the minimum cost accumulated for all algorithms. Since the algorithm was run a full round previously, $c_{A_i} \geq \gamma^{L-m}$ for all i, so $\frac{c_{\text{MinDet}}}{\min_j(c_{A_j})} \leq \frac{2\frac{\gamma^L}{\gamma-1}+c_{A_i}}{\min_j(c_{A_j})}$, coming from the prior bound yields $\frac{c_{\text{MinDet}}}{\min_j(c_{A_j})} \leq 2\frac{\gamma^m}{\gamma-1}$.

If $L \geq m$, but some other algorithm has the minimum cost then for that algorithm we find $\min_j(c_{A_j}) \geq \gamma^{L-m+1}$ as it must be at least one bound prior to end but still that $c_{A_j} \geq \gamma^L$ as otherwise L would increase. Thus, we follow $\frac{c_{\text{MinDet}}}{\min_j(c_{A_j})} \leq 2\frac{\gamma^{m-1}}{\gamma-1} + \gamma^{m-1}$ $leq2\frac{\gamma^m}{\gamma-1}$ coming again from the prior bound and dividing.

Between all cases, we therefore guarantee at least $(2\frac{\gamma^{|A|}}{\gamma-1} + 1) \cdot \min_i(c_{A_i})$ for this instance. □

The intuition for the construction of Algorithm 6.2 and subsequent Lemma 6.2 relies on the idea that cycling through increasingly cost-permitting algorithms allows sufficiently bad algorithms to be avoided, while still allowing some good performance. This guarantee at least provides robustness, once γ and m are declared.

Theorem 6.3. *Given a deterministic α-competitive online algorithm A for MTS problem M and action predictions for that problem, there is a deterministic algorithm achieving competitive ratio $9\min(\alpha, 1+4\frac{\eta}{c_{OFF}})$ against any offline algorithm OFF.*

Proof. Starting with Lemma 6.2, let $m = 2$ and $\gamma = \frac{m}{m-1}$. We therefore get a competitive ratio $2(m-1)(\frac{m}{m-1})^m + 1 = 9$ from MinDet against A. We make the selection of these two algorithms A and FTP. Since we can only make an α guarantee on A, we therefore have 9α robustness. However, should it be possible to use Lemma 6.1 as a better algorithm, we can instead guarantee off of that online algorithm a ratio of $9 \cdot (1 + 4\frac{\eta}{c_{\text{OFF}}})$.

Thus, we achieve the minimum of these with $9 \cdot \min(\alpha, 1 + 4\frac{\eta}{c_{\text{OFF}}})$. □

Importantly, this provides both 9α robustness and $9 \cdot (1 + \frac{4\eta}{c_{\text{OFF}}})$ as a guarantee which, for perfect advice, achieves a 9 consistency guarantee. We must make the observation that this guarantee is somewhat weak. $9 \cdot \alpha$ when starting from an α guarantee could, for a small problem, be particularly large. Although this is a competitive ratio on potentially optimal OFF and therefore related to the problem itself, this 9 term is not bound by diameter nor any other term of an optimal solution.

It is not unreasonable to consider a randomized scheme for a dynamic selection between a provided online algorithm and the predictor based on additional information gained. Moreover, [ACEPS23] finds that this can achieve strong results in its primary term, even allowing parameterized competitive relationship coming off of the provided algorithms and predictor error. However, doing so also includes a term where the competitive ratio becomes related to the problem instance diameter D. In some senses, this scheme is more successful in avoiding large factors on a provided algorithm for robustness, but this additional factor is rather inconvenient for blanket analysis.

Given in [ACEPS23] and originating in [BB00], Algorithm 6.3 seeks to be that randomized scheme. However, it requires some additional setup. We'll be using the concept of the transfer of probability mass as from the earth-mover distance of two probability schemes, intuitively conceived as the minimum distance-quantity product moved between probability mass functions. We will also notably need to know in advance the diameter of the metric space, that is the furthest distance between any two states in our MTS instance.

Algorithm 6.3: MinRand

Input: A parameter $0 < \varepsilon < 1/2$, set of online algorithms A with $|A| = m$, input sequence I for an MTS problem with diameter D

Procedure MinRand():
- $\beta \leftarrow 1 - \frac{\varepsilon}{2}$
- $w_i^0 \leftarrow 1 \ \forall i \in [0, m-1]$
- **for** *time* t *in* I **do**
  - c_i^t is the cost incurred by A_i at t
  - $w_i^{t+1} \leftarrow w_i^t \cdot \beta^{c_i^t/D}$
  - $p_i^{t+1} \leftarrow \frac{w_i^{t+1}}{\sum_{j\in[m]} w_j^{t+1}}$
  - $\tau_{i,j}$ is the probability mass transferred from p_i^t to p_j^{t+1} for two algorithms $i, j \in A$
  - Switch from A_i to A_j with probability $\frac{\tau_{i,j}}{p_i^t}$

The overall scheme in Algorithm 6.3 is to assign weights to each function and updates them related to a robustness-scaled factor on their cost yet incurred relative to one expectation, the diameter, on the cost in the problem instance. Then, it adjusts the probability functions for the new timestep (basically just recalculating relative probabilities as a softmax of their weights) and identifying that change as a transfer quantity τ. Then, swap to a new algorithm based on that transfer quantity, normalized into probabilities by the division of p_i^t. Note that MinRand is a WMR algorithm from [LW94] and following [BB00] rewritten for this problem. To be more cohesive with sources, we will first consider a WMR format for which $0 < \varepsilon_{WMR} < 1/4$, which allows an easier analysis, then rescale this value to make a cleaner cost bounds.

Lemma 6.4. *Where the best expert in a WMR has total loss L, the WMR at large incurs at most* $\frac{\ln(1/\beta)L + \ln m}{1-\beta}$

Proof. For the sake of brevity and for concision, we take this result from [LW94]. □

Then leveraging this we follow [BB00] to gather Lemma 6.5.

Lemma 6.5. *For WMR with initial probability distribution p, a loss following ℓ, and updates to form probability distribution p', then state distance $d(p, p') \leq \ln(1/\beta)p \cdot \ell$.*

Proof. Identify that $W' \leq W$ as loss can never increase weight totals.

For the weights w_i prior and w_i' following the trial, we gather their sums W and W' respectively such that $p_i = w_i/W$ and $p_i' = w_i'/W'$.

$d(p, p') = \sum_{i:p_i > p_i'} (\frac{w_i}{W} - \frac{w_i'}{W'}) \leq \sum_i (\frac{w_i}{W} - \frac{w_i'}{W'}) = \sum_i \frac{w_i}{W}(1 - \beta^{\ell_i})$

Then by properties of the WMR defining the relationship on probability transitions, we get that $d(p, p') \leq \sum_i p_i(\ell_i \ln(1/\beta))$ □

From these two components, we turn to make a bound on expected cost for MinRand specifically.

Lemma 6.6. *If the best state has total cost L and $\varepsilon = 1 - \beta \leq 1/4$, a WMR algorithm incurs an expected cost at most* $\frac{(1+\ln(1/\beta))(\ln(1/\beta)L + \ln m)}{1-\beta} \leq (1 + 2\varepsilon)L + (\frac{7}{6} + \frac{1}{\varepsilon}) \ln m.$

Proof. The WMR algorithm takes expected cost at timestep t at most $c_t \leq d(p^t, p^{t+1}) + p^t \ell^t$ for particular loss ℓ^t at time t.

Then, we leverage Lemma 6.5 to identify that this factor is $c_t \leq \ln(1/\beta)(p^t \ell^t) + p^t \ell^t$. Thus, by sum, our total expected cost is at most $(1 + \ln(1/\beta)) \cdot$ total expected loss

Then, we leverage Lemma 6.4 and the identification that our total expected loss is related to the best expert loss L, we gather the expected cost at most $\frac{(1+\ln(1/\beta))(\ln(1/\beta)L + \ln m)}{1-\beta}$.

Taking our values for ε, we leverage $\ln(1/\beta) = -\ln(1-\varepsilon) \leq \varepsilon + .61\varepsilon^2$. Then our term becomes at most $(1 + (\varepsilon + .61\varepsilon^2))((\varepsilon + .61\varepsilon^2)L + (1 + \varepsilon + .61\varepsilon^2)\ln m) \cdot \frac{1}{1-\beta} \leq ((1 + 2\varepsilon)L + (\frac{7}{6} + \frac{1}{\varepsilon})\ln m)\frac{1}{1-\beta}$ since we know that $\varepsilon < \frac{1}{4}$ by precondition. □

Note that there is some gap here, it is possible to, for any given legal ε, achieve a tighter expectation on this behaviour, however it generally fails to coalesece in to a bound as clean.

Now with an answer to expected costs WMR, we seek to consider the effect of the algorithms held within. Again, recall that MinRand is a constructed WMR.

Lemma 6.7. *Given m online algorithms in set A for a MTS with diameter at most $D > 0$ and a value $0 < \varepsilon < 1/2$, MinRand is an algorithm such that, for any instance I, its expected cost is at most $(1 + \varepsilon) \cdot \min_t(c_{A_i}) + O(D/\varepsilon) \cdot \ln m$.*

Proof. We will make the adaptation that $2\varepsilon' = 2\varepsilon_{WMR} = \varepsilon_{\text{MinRand}} = \varepsilon$ for convenience and cohesion with [ACEPS23].

First rescale so that any cost vector ℓ is normalized to $\ell' = \frac{\ell}{D}$ and similarly $L' = L/D = \min_t(c_{A_i})/D$ to permit Lemma 6.6. We then leverage that lemma that so the cost of the function at large becomes $(1 + \varepsilon)\min_t(c_{A_i})/D + (\frac{7}{6} + \frac{1}{2\varepsilon})\ln m$.

To rescale these, multiply back D such that we now observe $(1 + \varepsilon)\min_t(c_{A_i}) + O(D/\varepsilon)\ln(m)$. □

So we can follow the same basic structure as in the determinant case to gather out final conclusion.

Theorem 6.8. *Given a randomized α-competitive online algorithm A for MTS problem M with metric diameter D, a parameter $0 < \varepsilon \leq 1/4$, and action predictions for that problem, there is a randomized algorithm expected to achieve cost at worst $(1 + \varepsilon) \cdot \min(\alpha, 1 + 4\frac{\eta}{c_{OFF}}) \cdot c_{OFF} + O(D/\varepsilon)$.*

Proof. Starting with Lemma 6.7, let $m = 2$ and make the selection of these two algorithms A and FTP with guarantee from Lemma 6.1. We therefore get a cost ratio upper bound of $(1+\varepsilon)\cdot\min_i(\alpha, 1+4\frac{\eta}{\text{OFF}}) + O(D/\varepsilon)$ by using at-least-the-worse guarantee of these two algorithms. □

It would be remiss as addendum to fail to mention [SE24] in this context. Unlike some of the previous problems, algorithms to *MTS* involve a large number of predictions. As is perhaps expected, the quantity of these predictions and the total information carried by them is also relevant to the design of an algorithm. Although we will consider that discussion out of the scope of this particular problem, there is some value in identifying that the quantity, and not merely quality, of predictions has implicitly influenced *all* of the algorithms surveyed.

For this construction, however, there is an evident variation in algorithmic design which tends toward a distinct intuition. In the deterministic approach, setting what are effectively tolerances or budgets for bad behaviour as an algorithm progresses are made dynamic. The algorithm begins with a budget for cost and is granted more only when it and all other algorithms exceed their budgets. However, this still achieves fairly immediate robustness guarantees. On the other hand, the randomized algorithm manages to tighten this robustness guarantee in some manners by adapting this switch probabilistically between costs yet experienced. Effectively, this leverages more classical explore-exploit concepts to achieve bounds on expected cost.

7 Conclusion

Generally, these learning-augmented approaches have provided a new avenue to leverage both growing tools in applications and to explore classical problems with new expectations. Seeking to loosen online problems expectations about what information is available to an algorithm has provided all of the problems discussed here some capability for results which are potentially far better than traditional online approaches. Moreover, with our considerations of robustness, for many of these problems, we still achieve strong worst-case guarantees.

We see already a number of interesting sub-categories. In *MTS* and *Online Metric Facility Location*, algorithms rely on integrating predictions into the decision making, dynamically avoiding worst-case predictions while trying to leverage plausible predictions on every step. In *Ski Rental*, *Online Set Cover*, and *MTS*, instead algorithms choose to create a set of strong opening predictions which guide performance until the algorithm gets 'too bad'. Along the same lines, we see a distinction that some algorithms involve a large number of predictions, whereas others fundamentally avoid needing, or leveraging, repeat predictions. Moreover results are generally mixed. For some, achieving 1-consistency and seeking to raise robustness is reasonable, but generally there exists an active tradeoff between robustness and possible consistency for the problem.

In light of these approaches and in the growth of the field, seeking to integrate 'plausibly correct' information is a somewhat natural extension of bridging the usability gap of strongly-correct information calculable via inputs and possibly-(or expectedly-)correct random guidance in solving these online problems.

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